Online Pre-Work:
Reproductive System Embryology

Elizabeth A. Spudich, Ph.D.

JeffMD

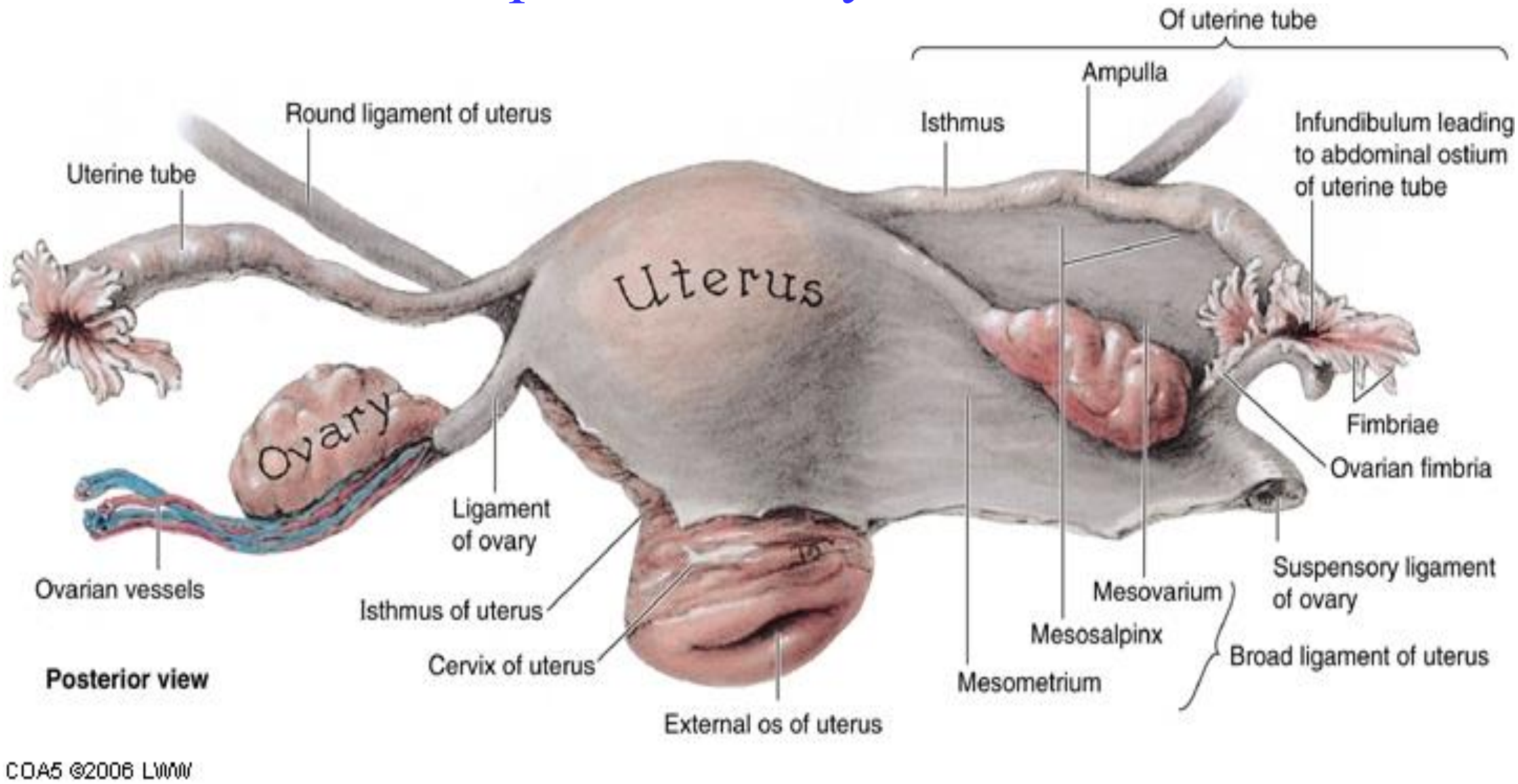
Sidney Kimmel Medical College

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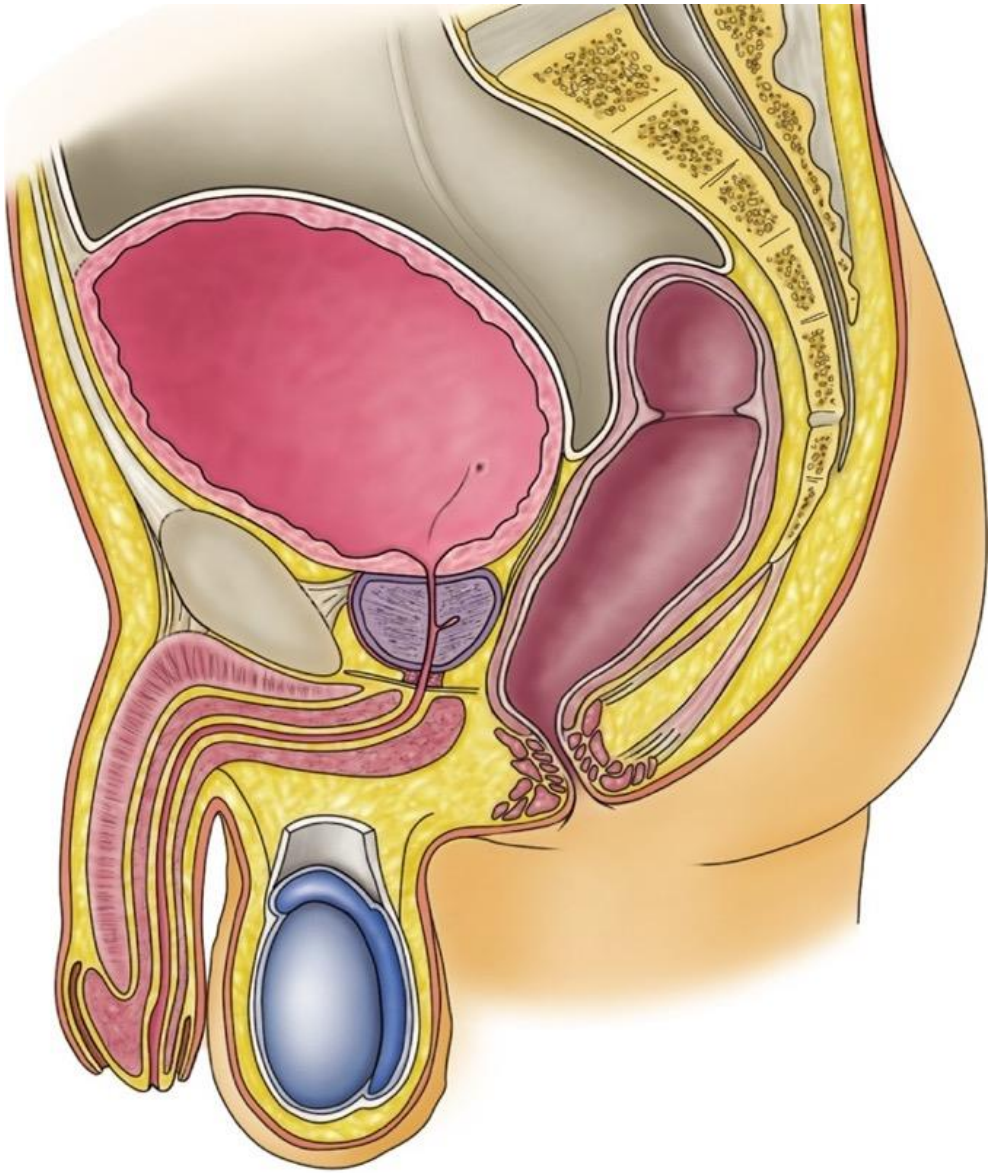
Reproductive System Learning Objectives

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Reproductive System

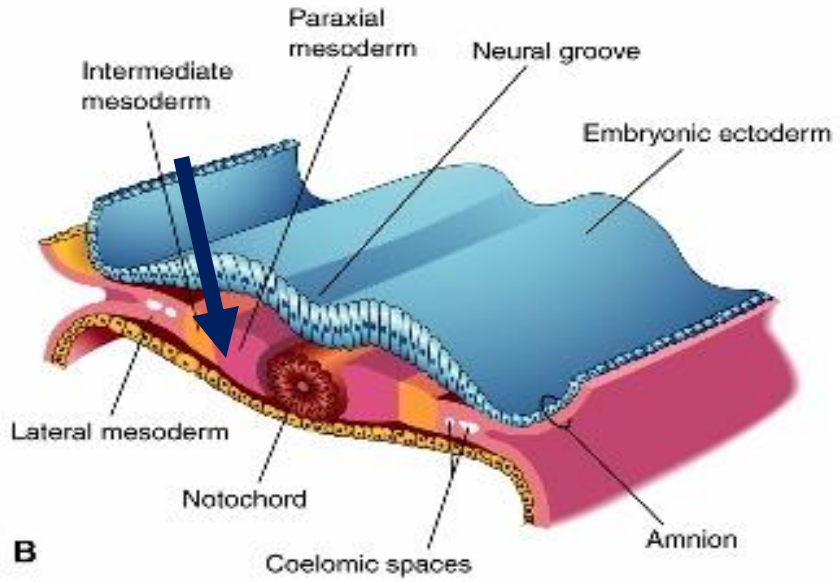


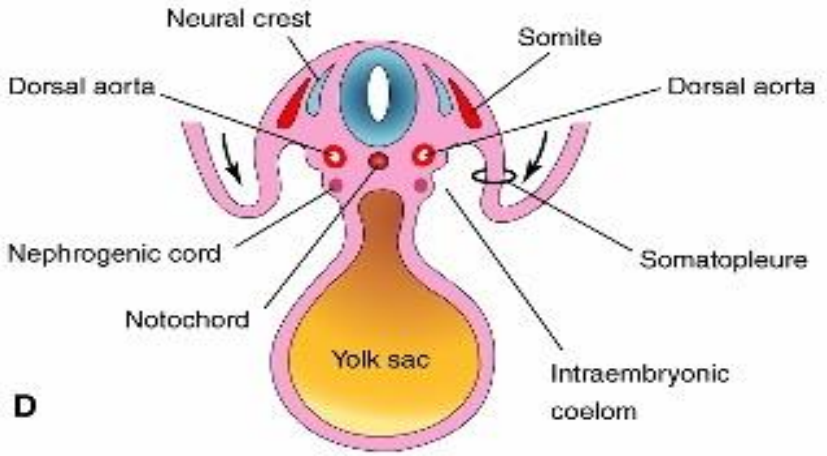
Reproductive System



Urogenital System

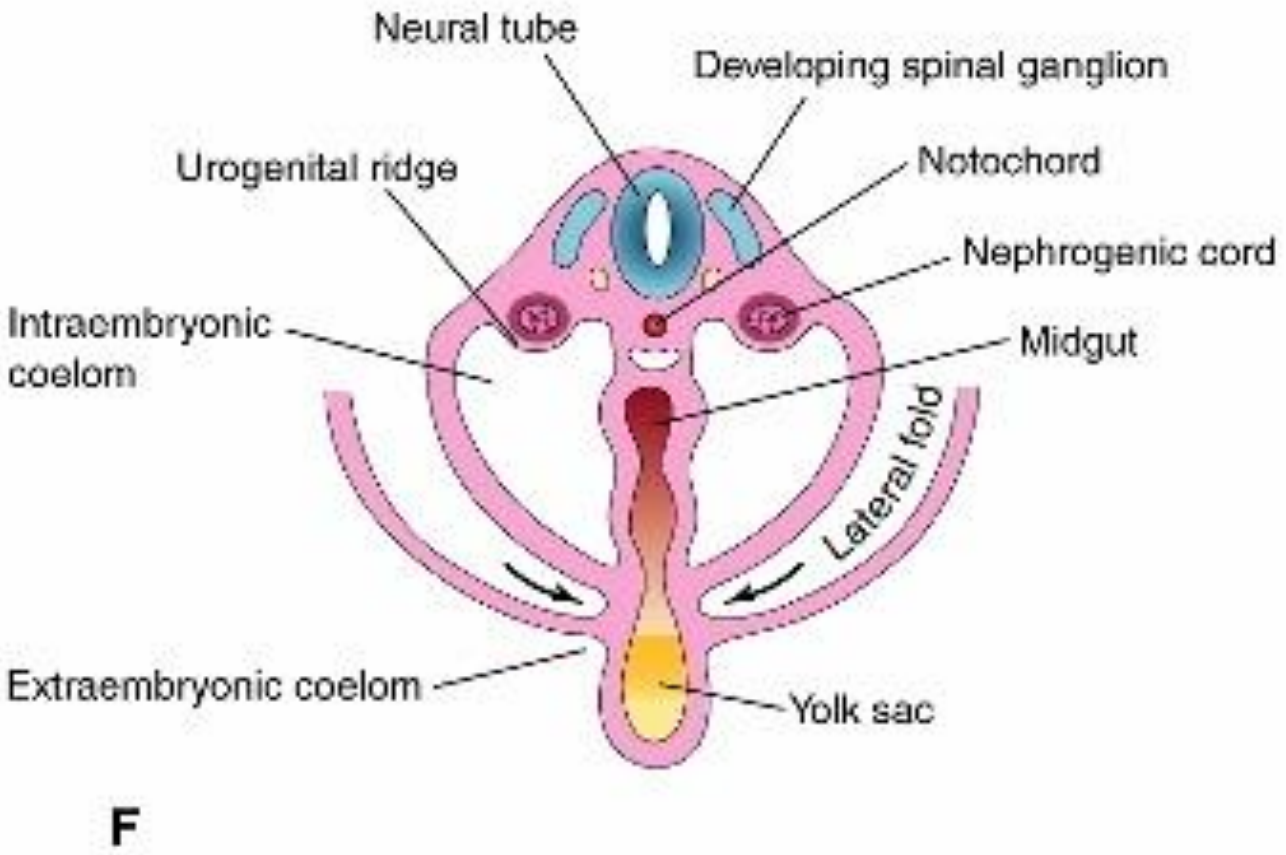
- Derived from the Intermediate Mesoderm
  - Connection to the presumptive somites
- During folding this mesenchyme pulled ventrally





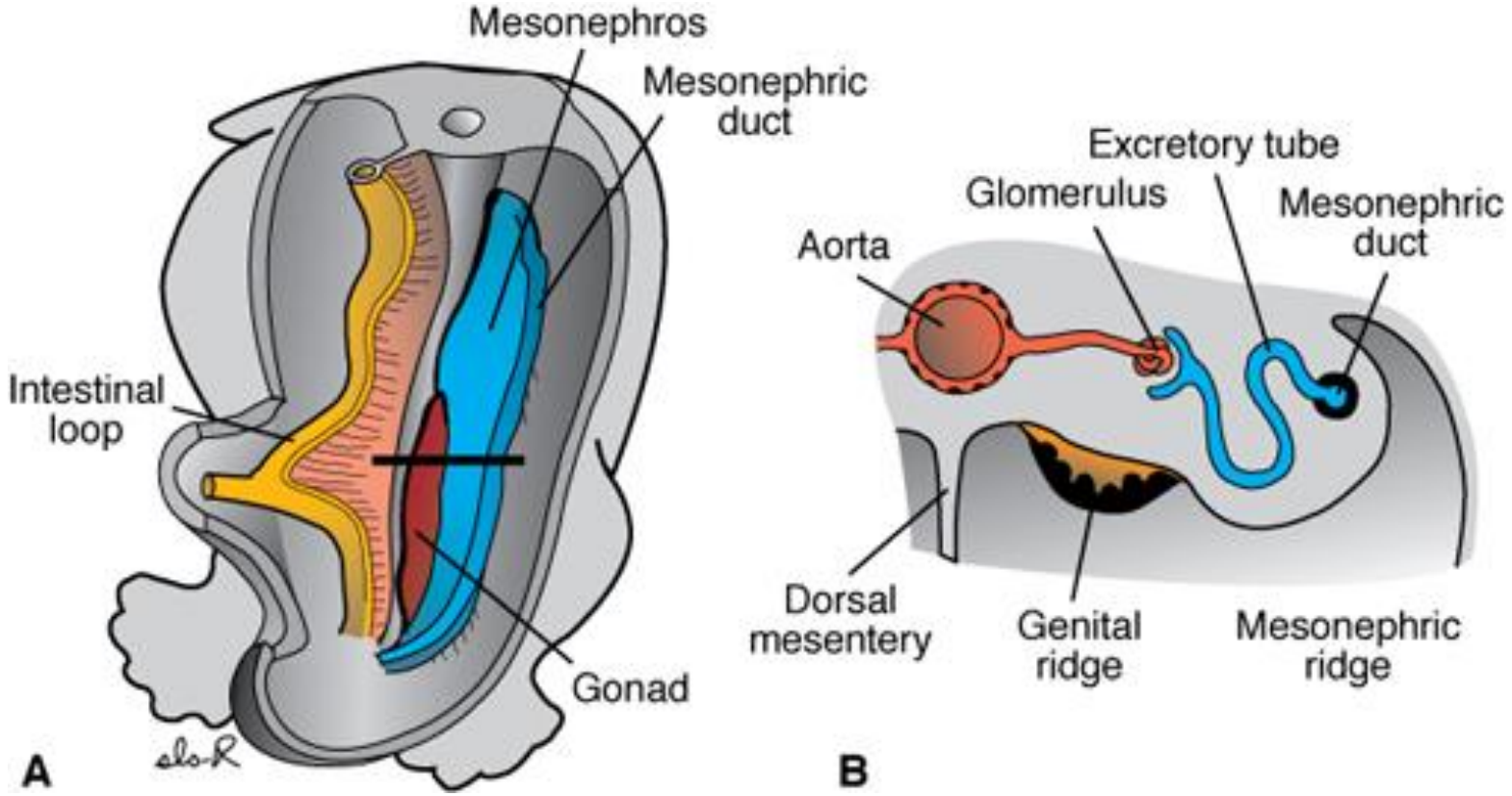
Urogenital System

- At closure of the body wall tissue forms a urogenital ridge along the dorsal body wall
- Two components to ridge
  - Nephrogenic cord
    - Extends from the cervical somites to the cloaca
  - Gonadal ridge



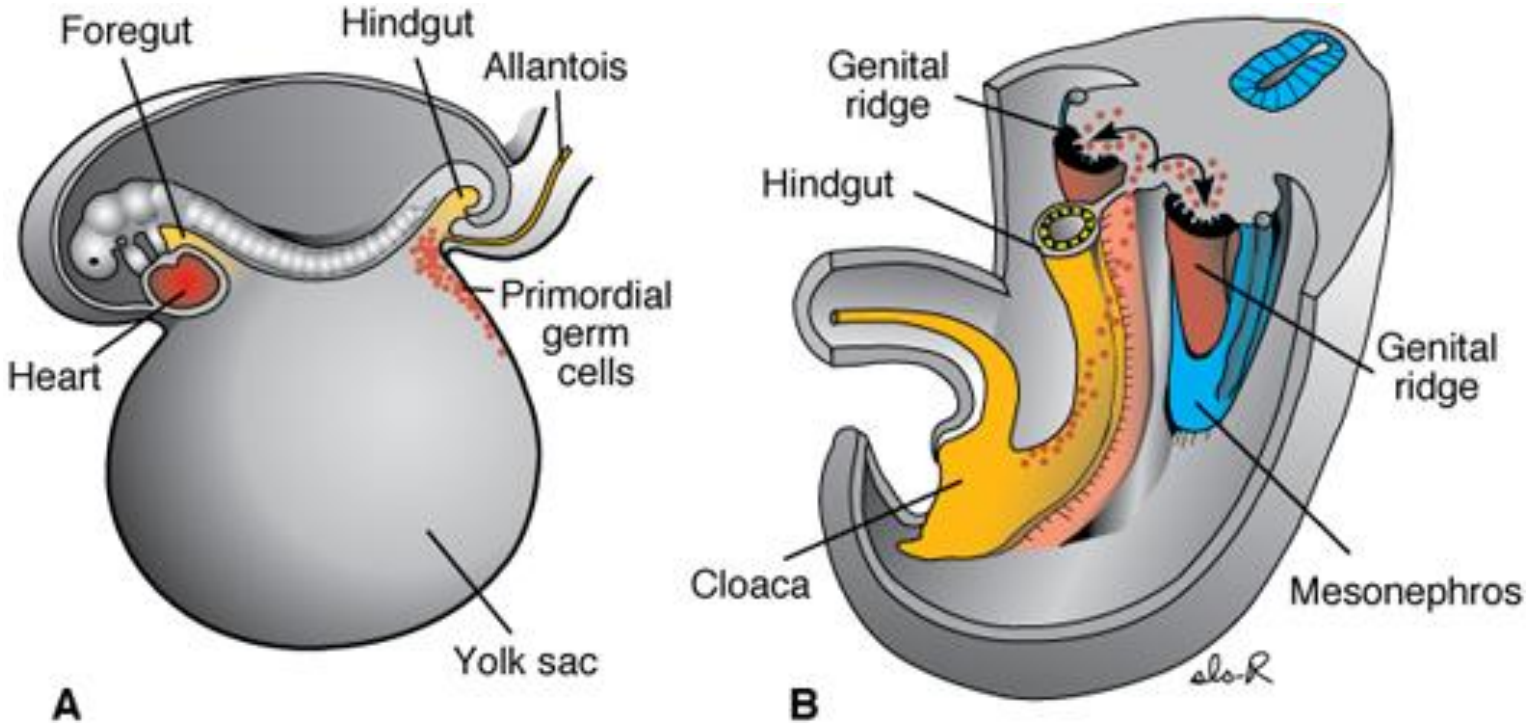
Genital Ridge

- Development of gonads start the same in everyone
- Week 5 there is a thickening of the tissue ventral to mesonephric ridge
  - Genital ridge



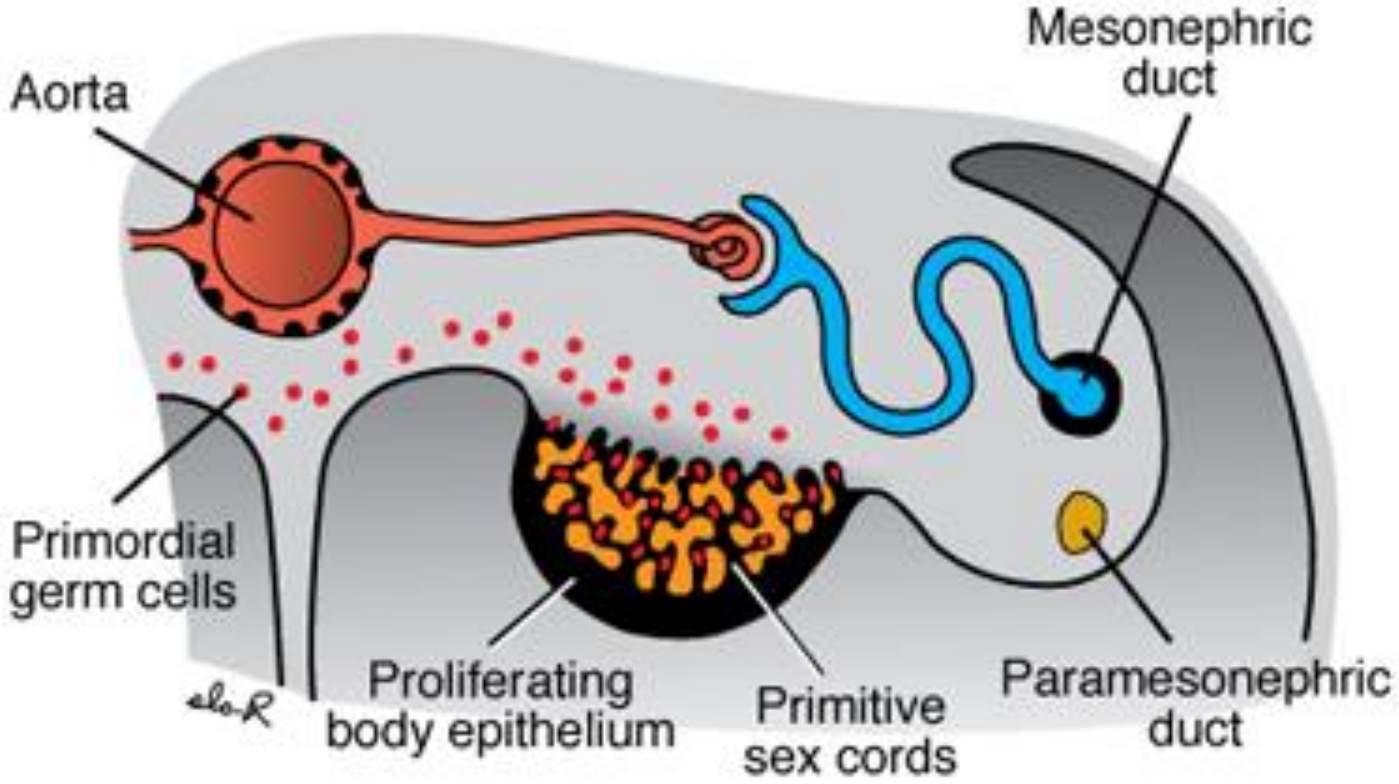
Germ Cells

- Germ cells arise from the epiblast layer early in development
  - Move into the yolk sac very early in the developmental time frame
- Start migration BACK to seed the genital ridge in week 4



Gonadal Cords

- Epithelial cells on ridge proliferate to form gonadal chords
  - Subdivided to cortex and medulla
- Cords are populated by germ cells in week 6

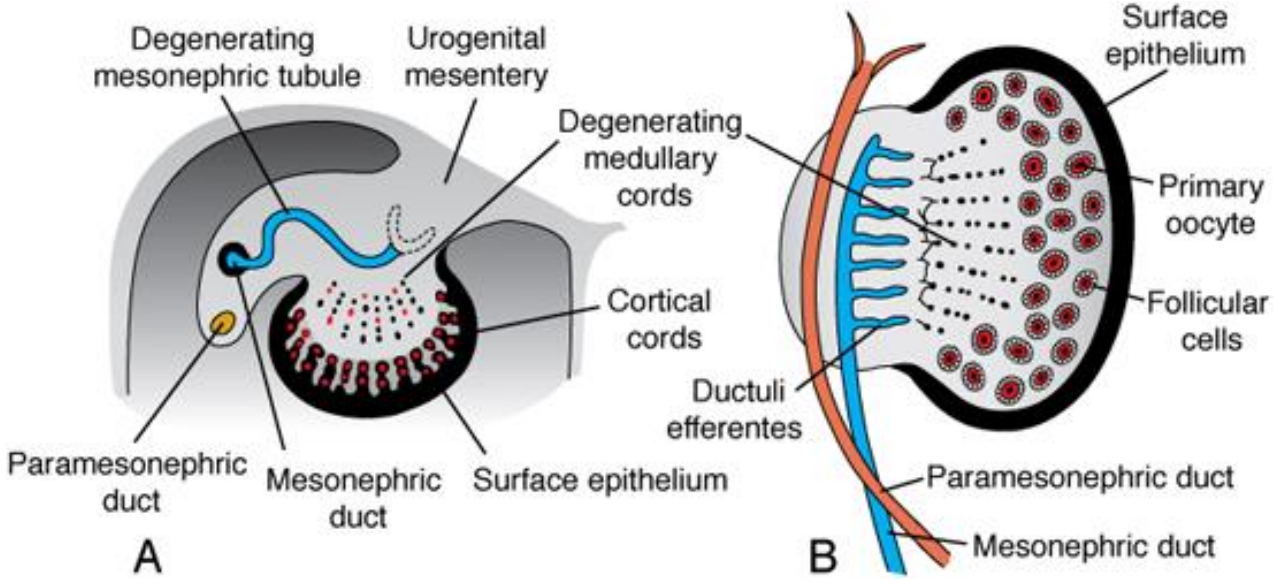


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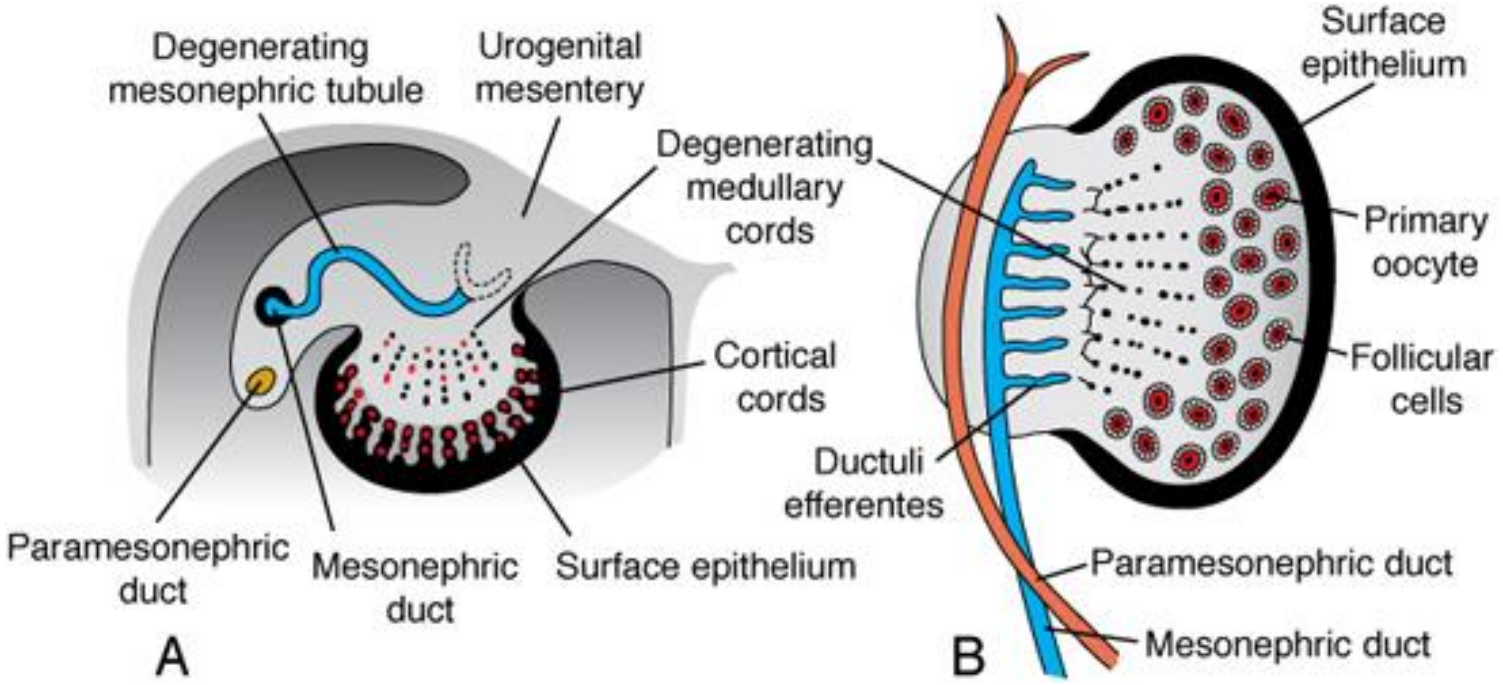
X Chromosome-Driven Gonadal Differentiation

- Differentiation in the absence of Y chromosome
- Primitive gonadal cords bunch up as clusters and persist at the cortical region
  - Remain proliferative
- Medulla becomes a vascular stroma



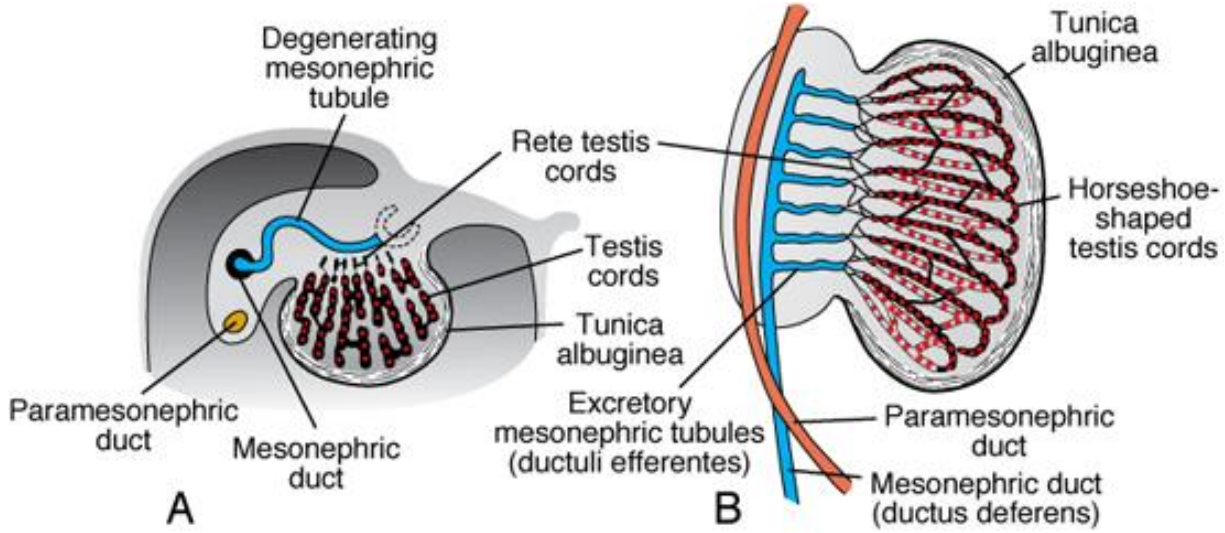
X Chromosome-Driven Gonadal Differentiation

- Gonadal cord cells surround each primitive oogonia
  - Follicular cells
- Cells from the Cord plus oogonium make the primordial follicles



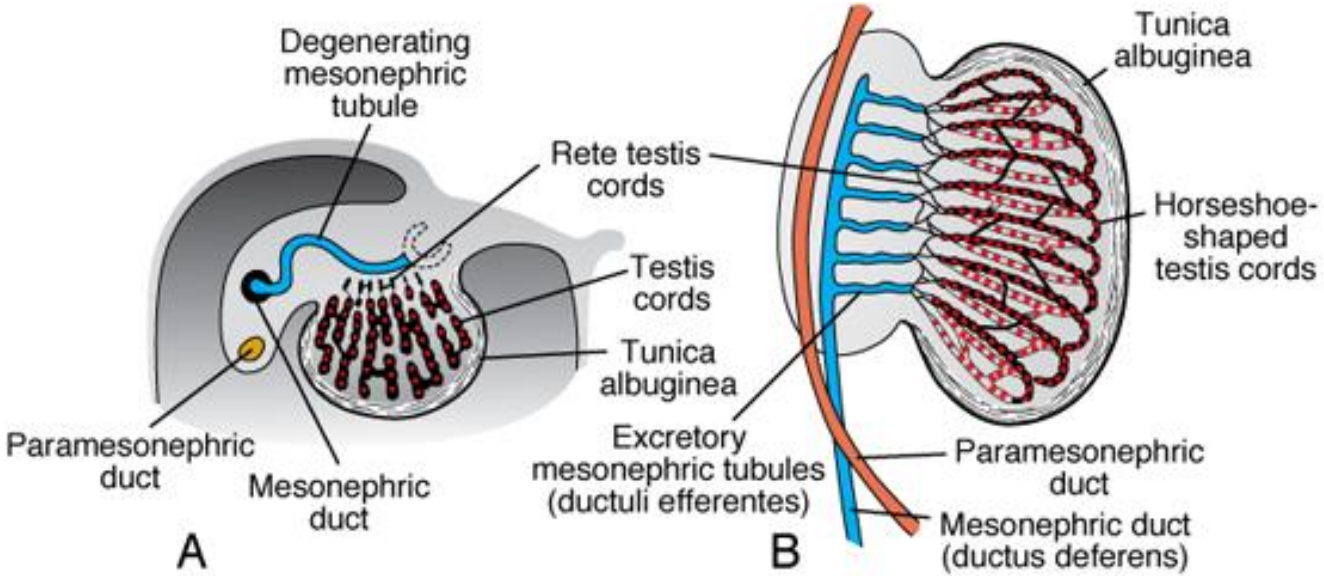
Y Chromosome-Driven Gonadal Differentiation

- The Y-chromosome drives the maturation of the testis
- Testis Determining Factor (SRY gene)
- Primitive gonadal cords extend to medullary region as solid testis cords and rete testis.
  - Encasement in Tunica albuginea
  - Solid cords becomes tubes during puberty

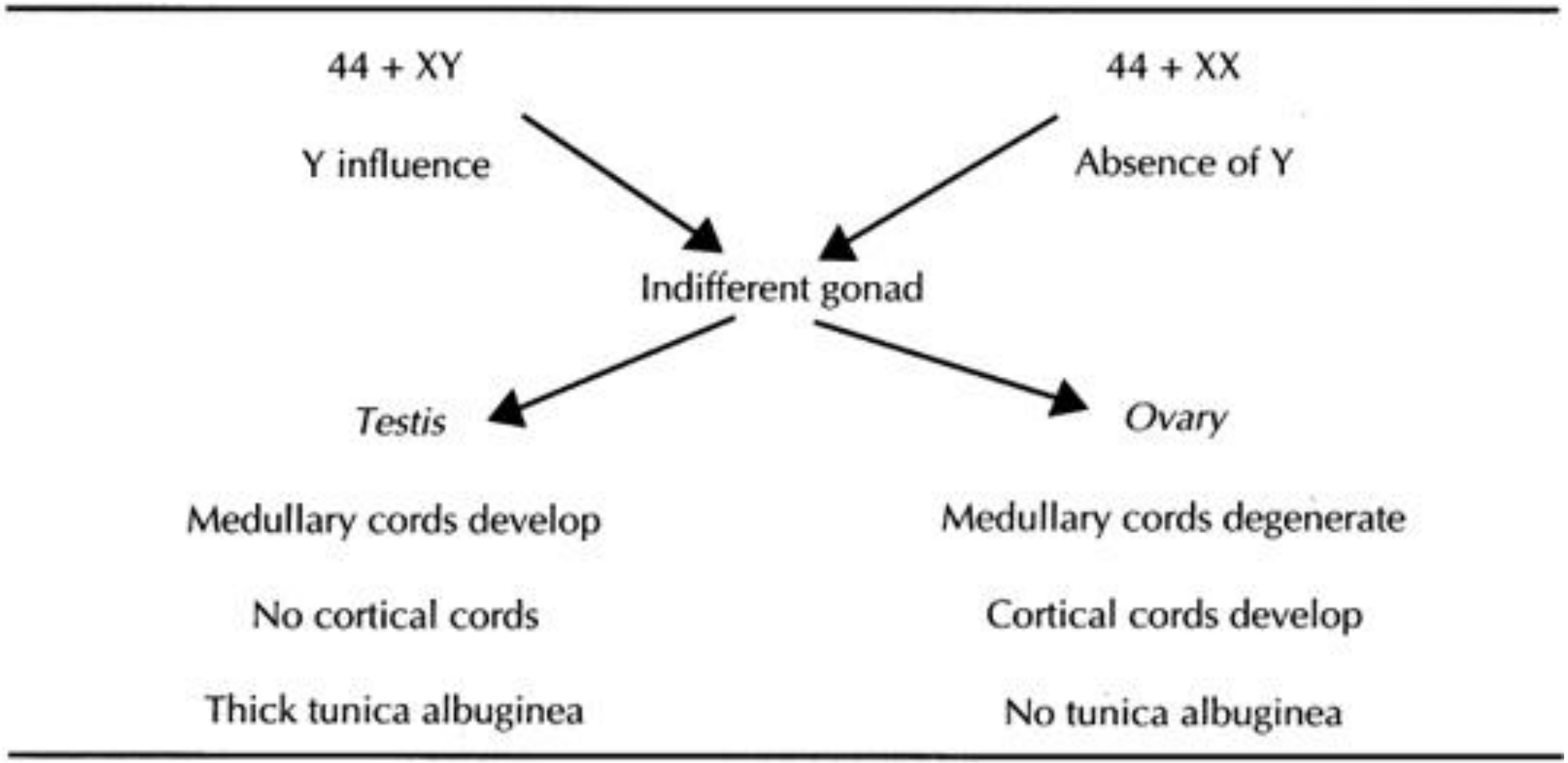


Y Chromosome-Driven Gonadal Differentiation

- Sertoli cells
  - Support cells from the surface epithelium of the primordial genital ridge that became cords
- Leydig cells
  - Make testosterone by Week 8
  - From the mesenchyme between the cords



Summary

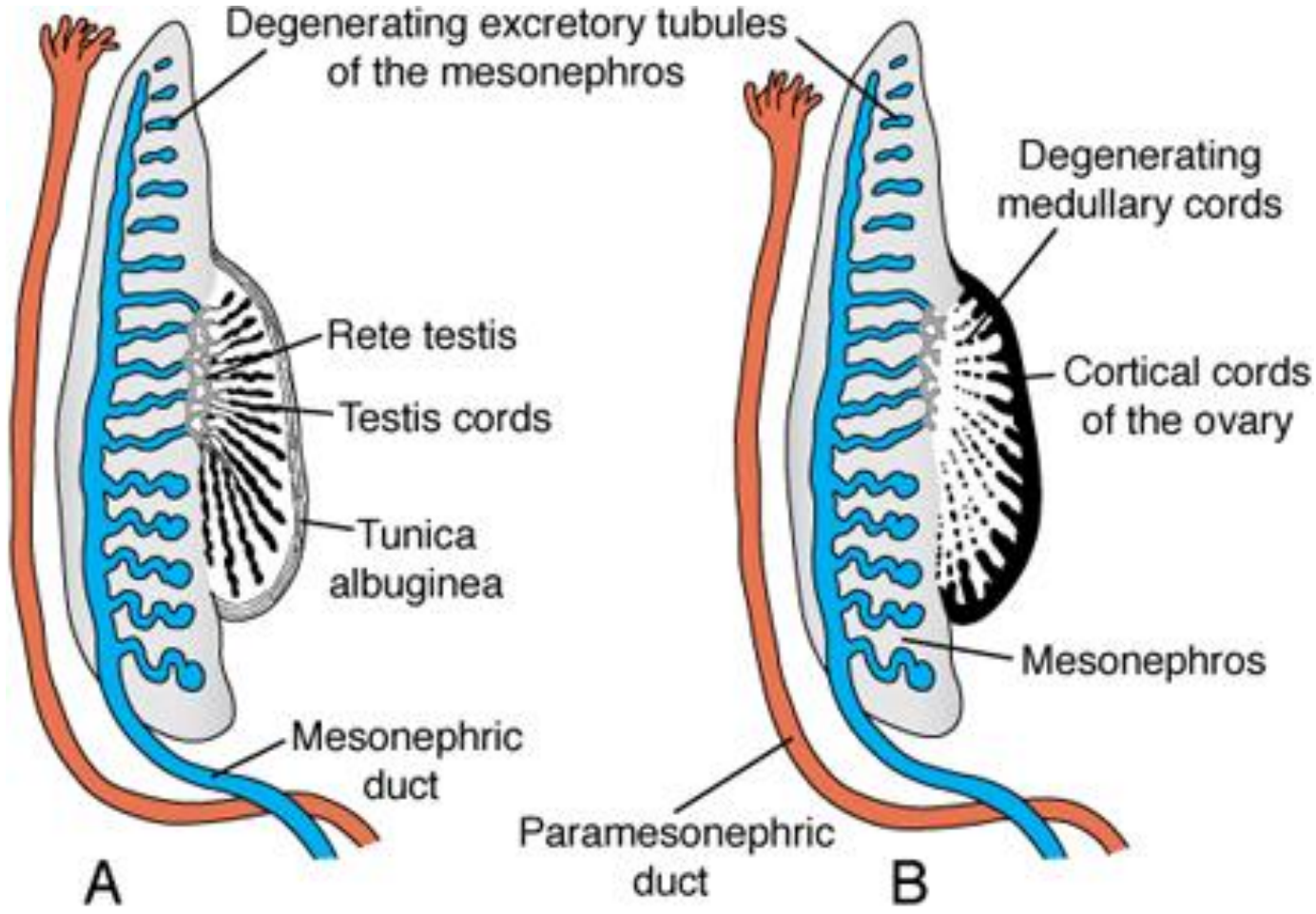

44 + XY
44 + XX
Y influence
Absence of Y
Indifferent gonad
Testis
Ovary
Medullary cords develop
Medullary cords degenerate
No cortical cords
Cortical cords develop
Thick tunica albuginea
No tunica albuginea

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Genital Duct Systems – Indifferent Phase

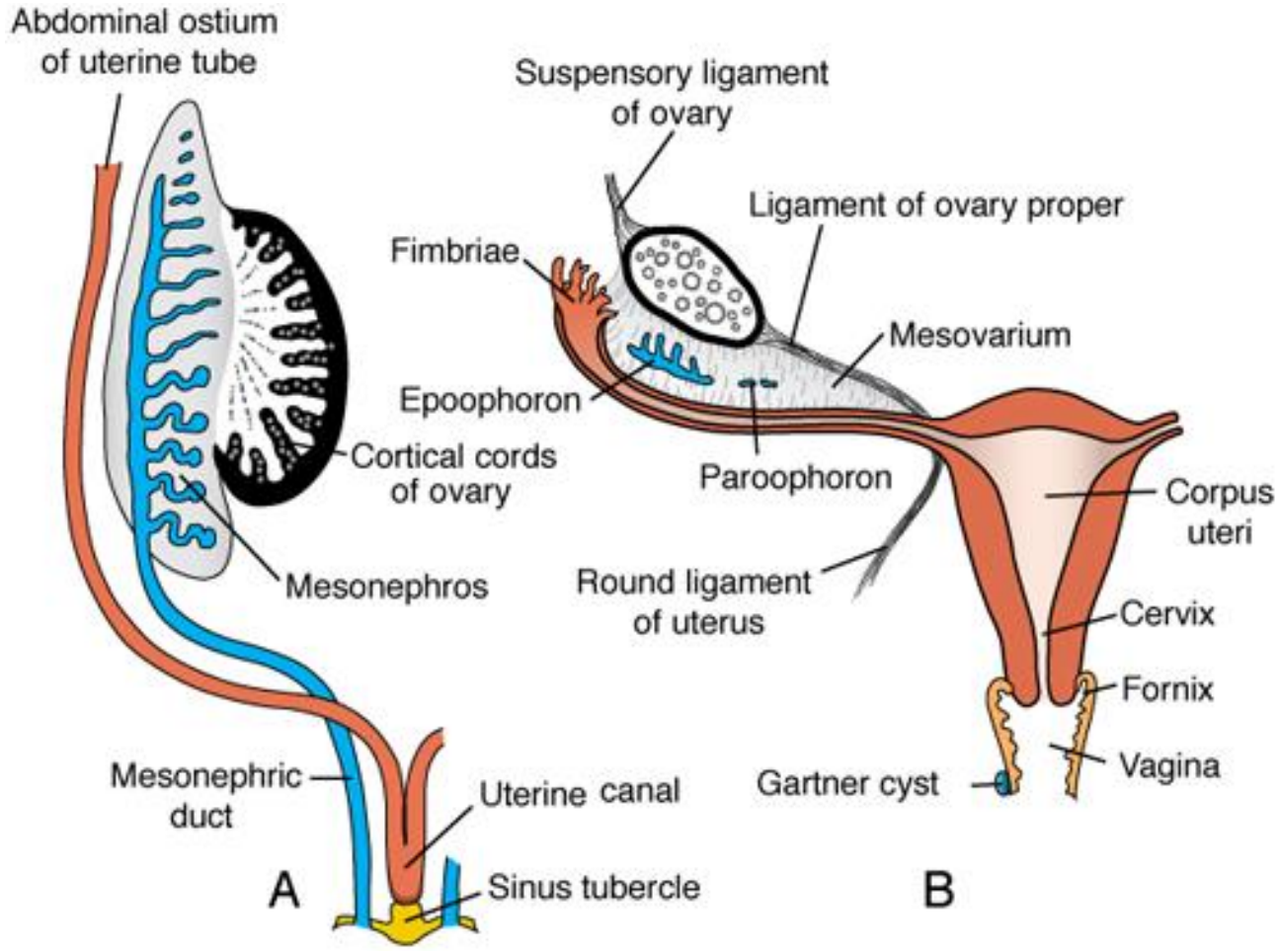
- Everyone starts with two ductal systems
- Mesonephric ducts of the intermediate renal system
  - a.k.a. Wolffian ducts
- Paramesonephric duct
  - a.k.a. Mullerian ducts



Paramesonephric Duct Development

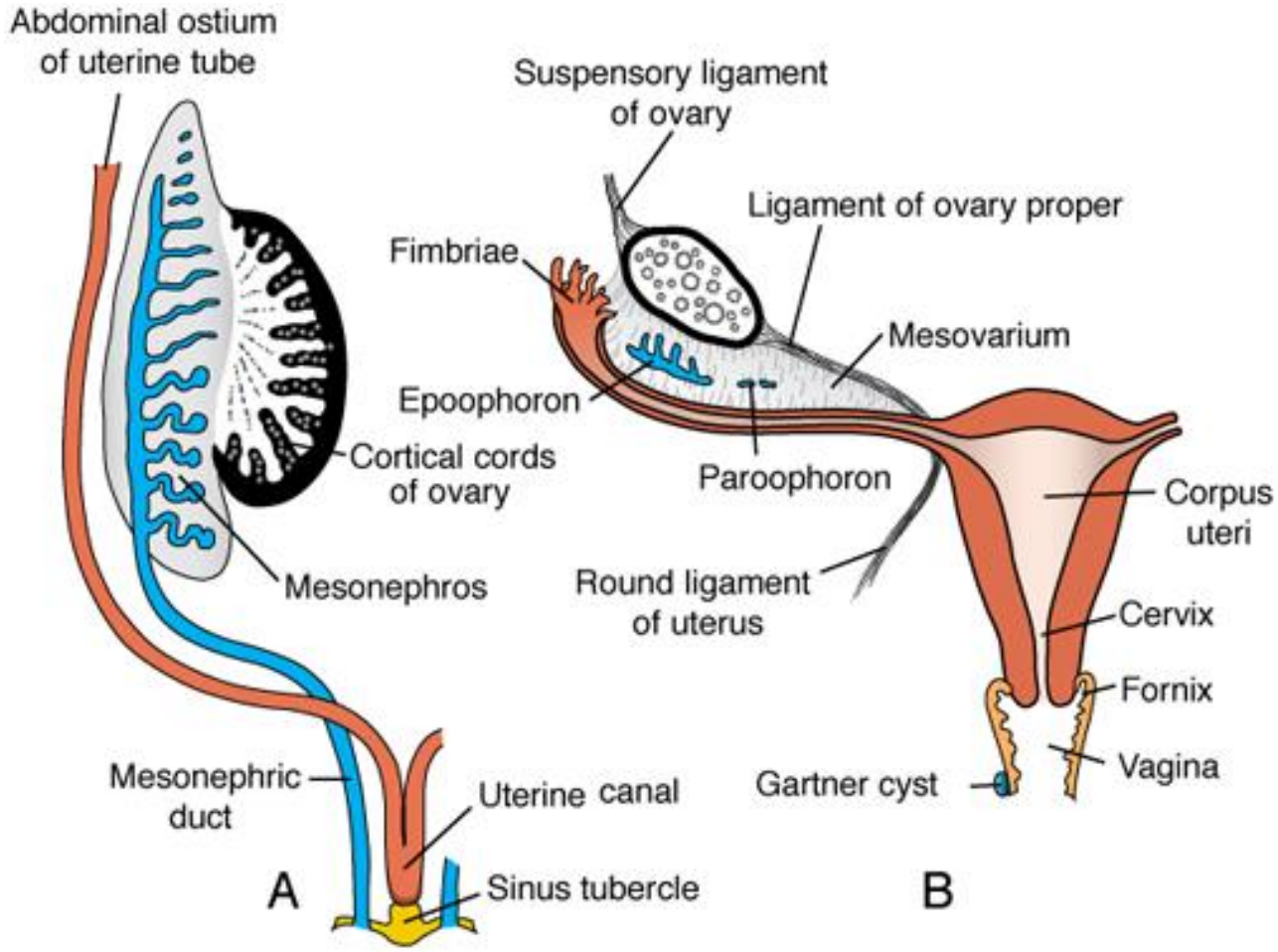
- Arises from an epithelial tissue lateral to genital ridge
- Three regions

1. Open to abdominal cavity cranially
   - Runs lateral to mesonephric duct
2. Turns medial and crosses mesonephric
3. Caudal end
   - Close to its opposite paramesonephric duct
   - Both fuse insert to the urogenital sinus



Paramesonephric Duct Derivatives

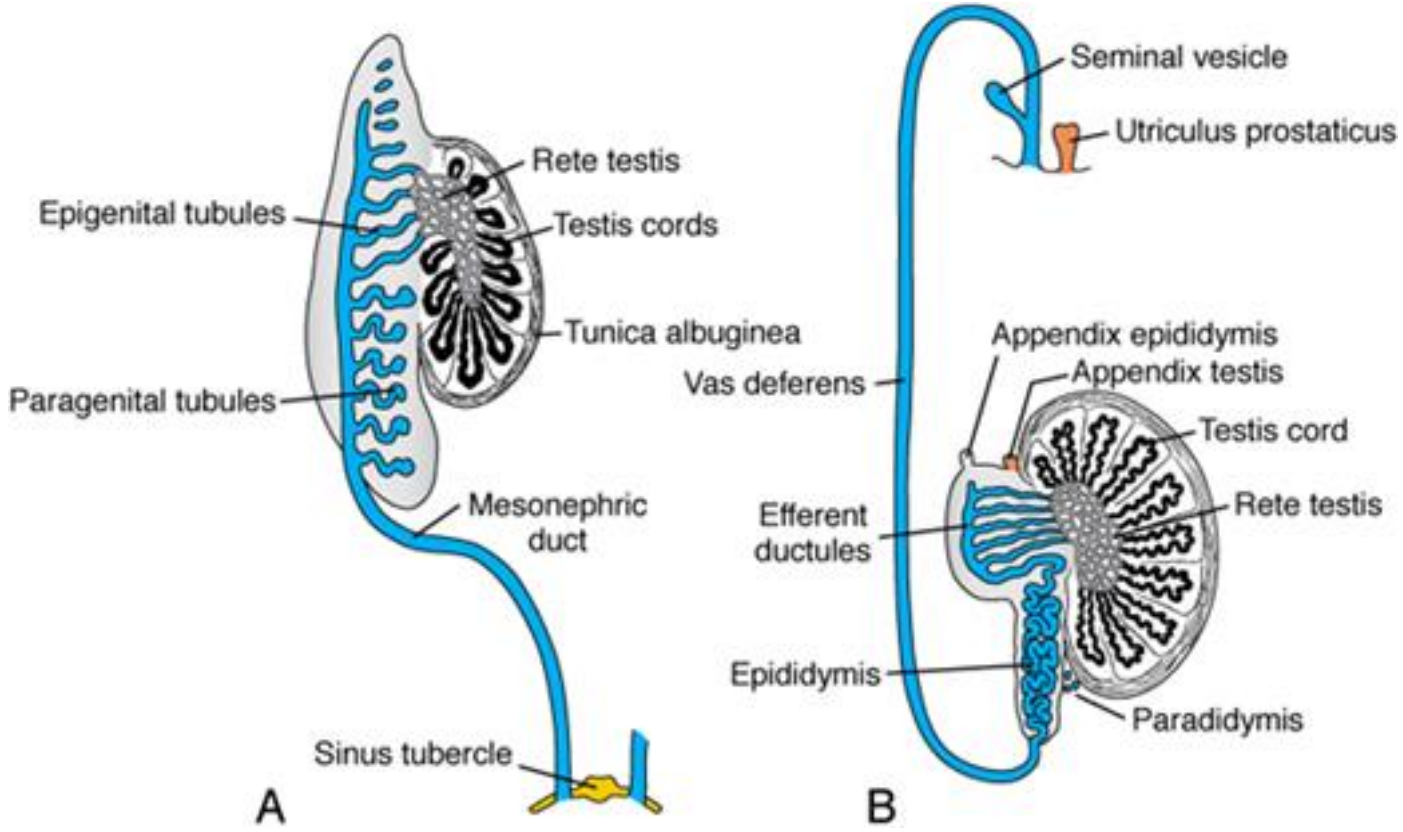
- Develops in the presence of estrogen AND the absence of testosterone and MIS/AMH
- Cranial aspects become the fimbriae of the uterine tubes
- Middle region becomes the bulk of the fallopian tubes
- Caudal aspect fuses to form the uterine canal
- Vaginal cavity has two origins
  - Upper from uterine canal
  - Lower from urogenital sinus (cloaca)



Mesonephric Duct Derivatives

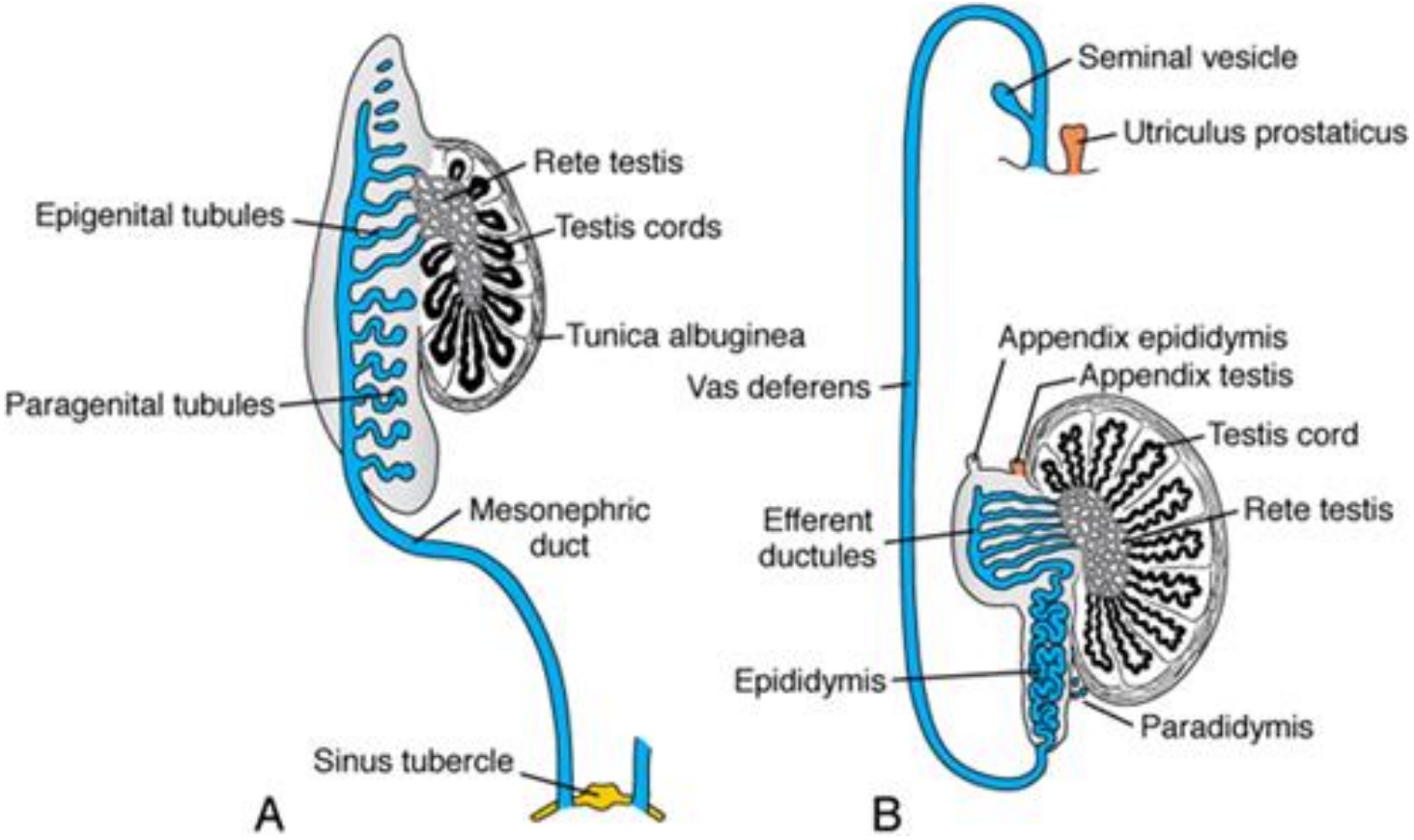
- Requires testosterone
- Mesonephric ducts remain

1. Cranial becomes the epididymis
   - Uses some of original renal tubules
2. Middle becomes the vas deferens
   - Seminal vesicles sprout near urogenital sinus
3. Caudal to the seminal vesicles becomes the ejaculatory duct

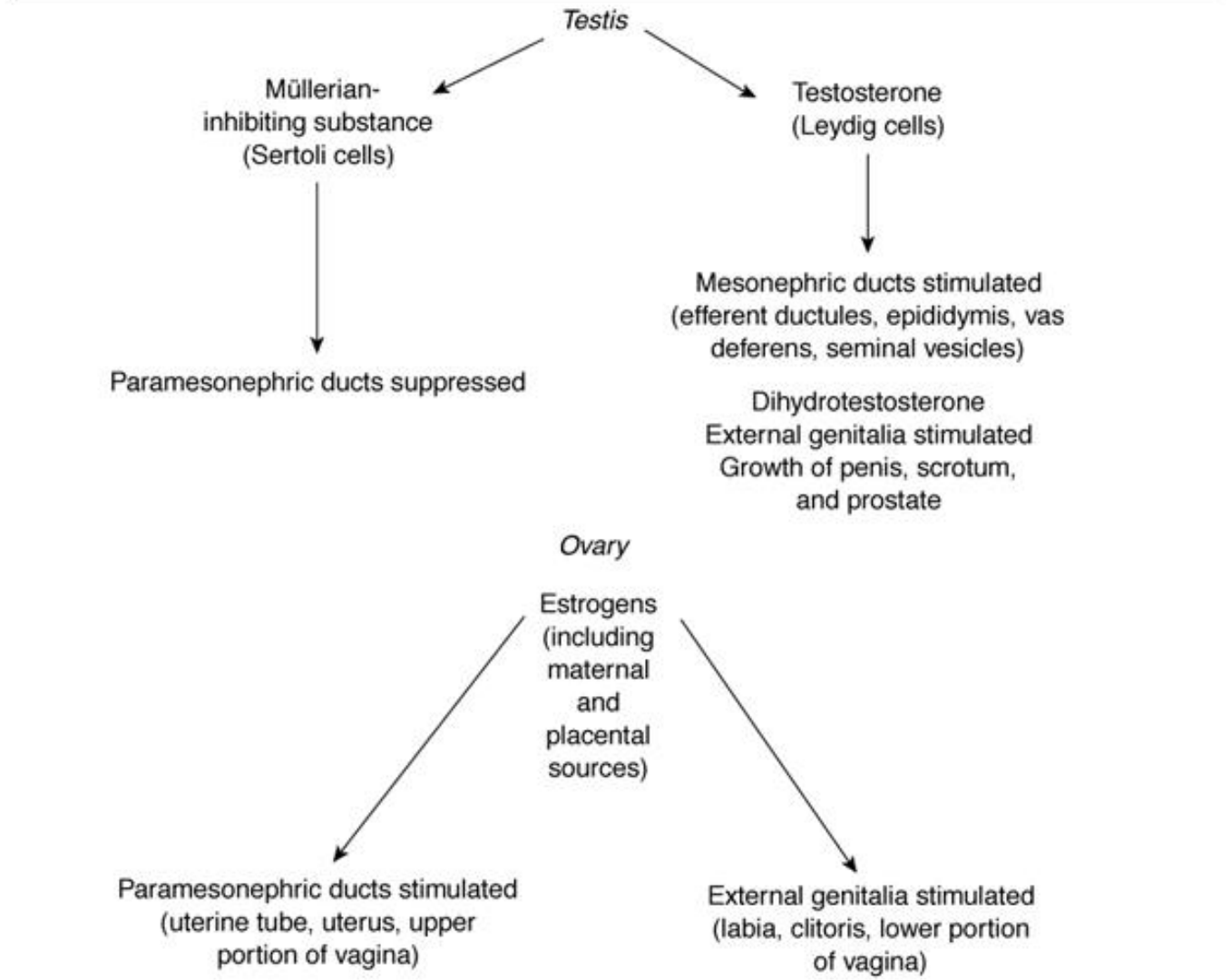


Mesonephric Duct Development

- Testosterone drives secretion of Anti-Mullerian Hormone (AMH)
  - AMH Secreted from Sertoli cells
  - Causes the degeneration of Paramesonephric ducts
    - Appendix of the testis is an aberrant remnant of the paramesonephric ductal system and can undergo torsion

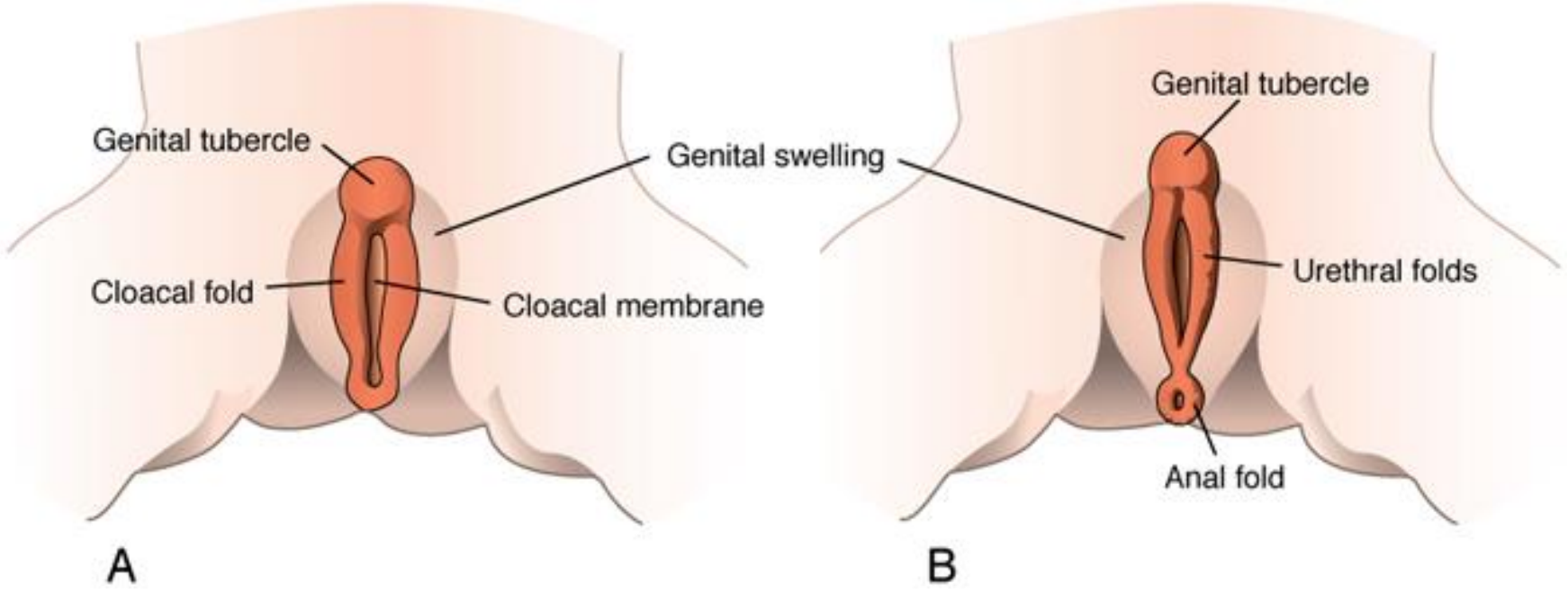


Summary


Testis
Müllerian-inhibiting substance (Sertoli cells)
Testosterone (Leydig cells)
Paramesonephric ducts suppressed
Mesonephric ducts stimulated (efferent ductules, epididymis, vas deferens, seminal vesicles)
Dihydrotestosterone
External genitalia stimulated
Growth of penis, scrotum, and prostate
Ovary
Estrogens (including maternal and placental sources)
Paramesonephric ducts stimulated (uterine tube, uterus, upper portion of vagina)
External genitalia stimulated (labia, clitoris, lower portion of vagina)

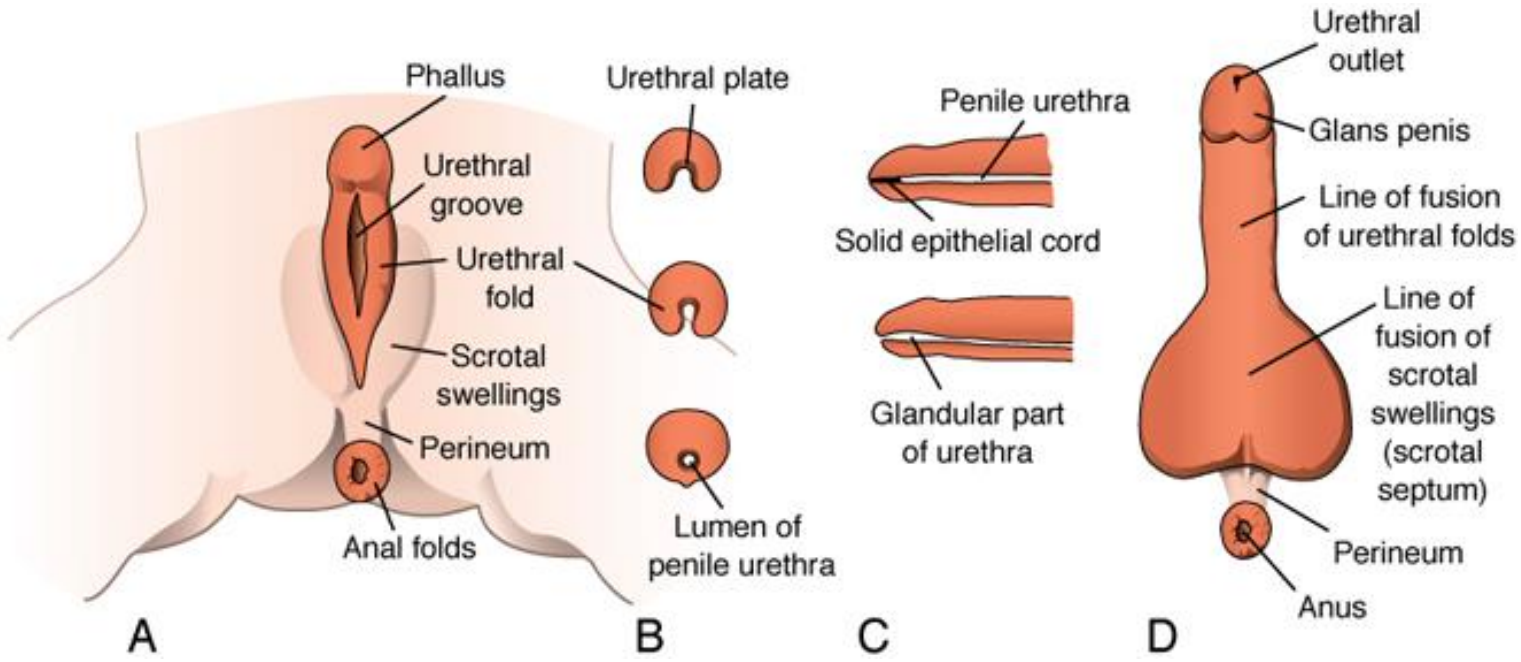
External Genitalia – Indifferent Phase

- Mesenchymal cells migrate to the cloacal opening in 3rd week as cloacal folds
- By fourth week
  1. Folds fuse cranial to membrane as the genital tubercle
  2. Urethral folds laterally
  3. Caudal regions fuse to give the anal folds
- Additional genital swellings lateral to the urethral folds



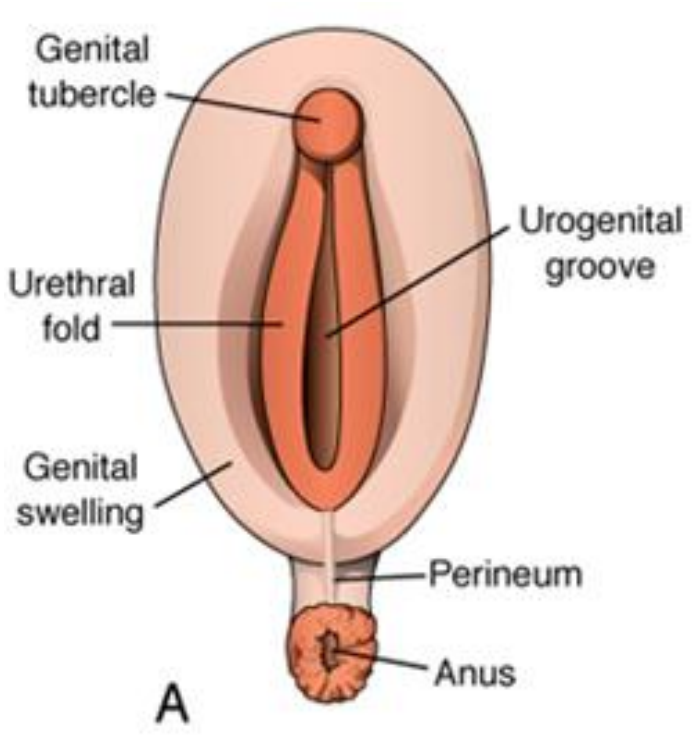
External Genitalia – Testosterone Driven Development

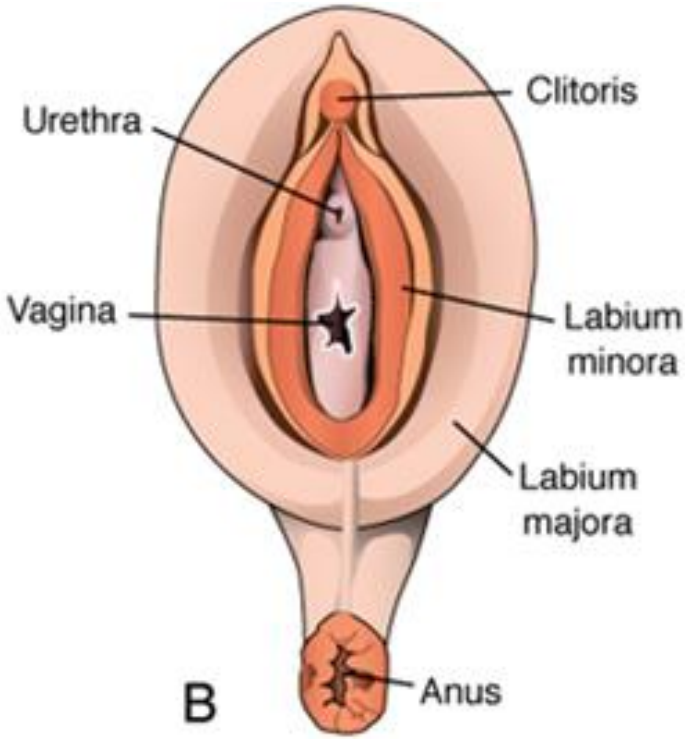
- Elongation of genital tubercle into phallus
- Urethral folds wrap around and fuse ventrally
- Endodermal cells of cloaca form lining called the urethral plate
  - By month three the folds enclose the plate
- Genital swellings = scrotal swellings



External Genitalia – Estrogen Driven Development

- Genital tubercle gets smaller and becomes clitoris
- Urethral folds don't fuse
  - Become the labia minora
- Urogenital groove becomes the vestibule of the vagina
- Genital swellings = labia majora



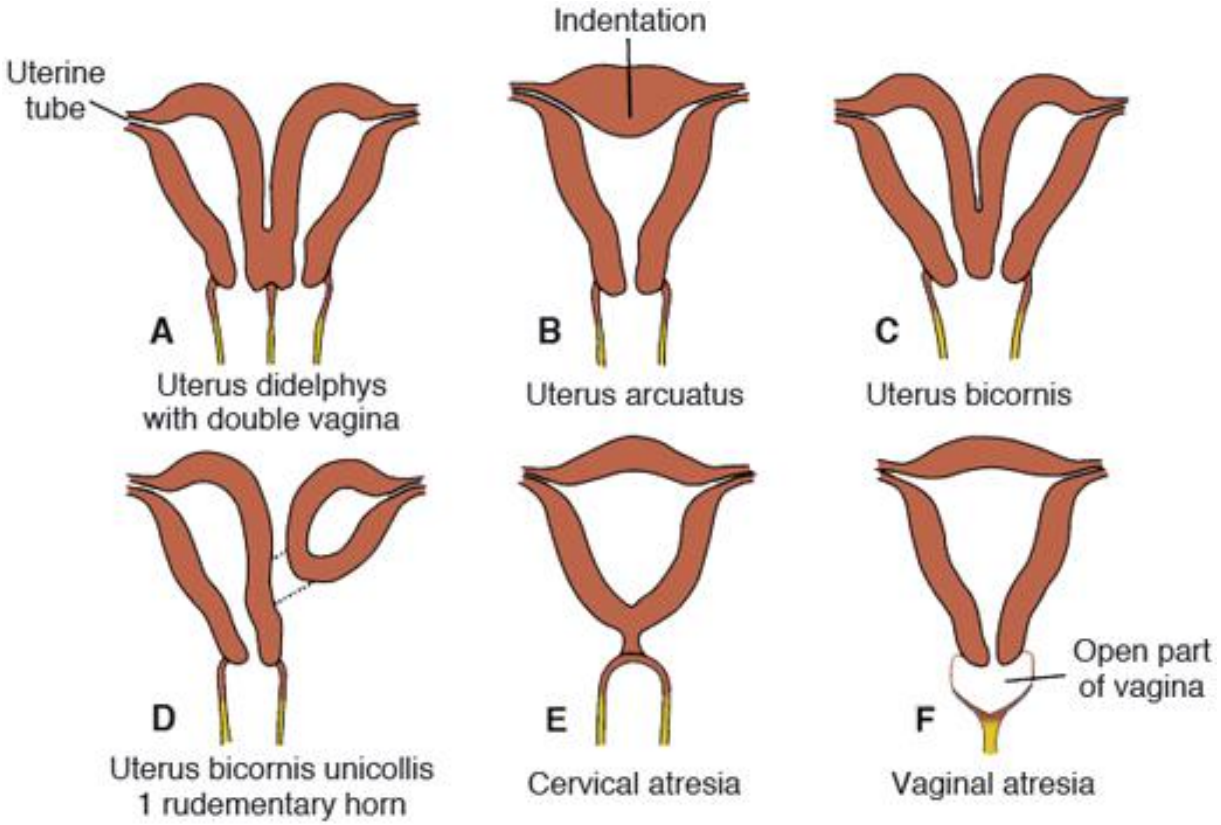


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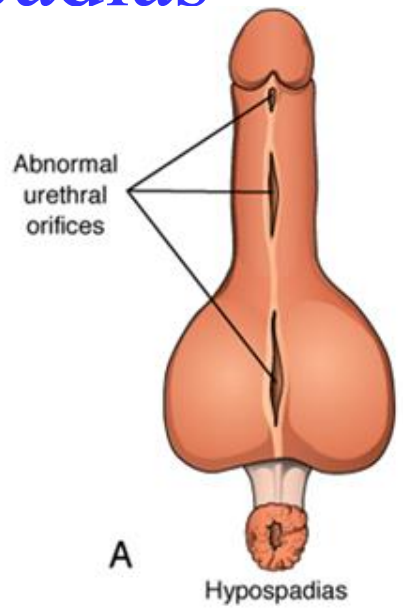
Uterine Structural Differences

- Abnormal fusions of the paramesonephric ducts
- DES exposure *in utero*
  - Diethylstilbestrol
  - Synthetic estrogen
  - Prevent miscarriages
  - 1938-1971
- DES daughters
  - Primary infertility because of structural changes
  - Surgical repair or assisted reproduction



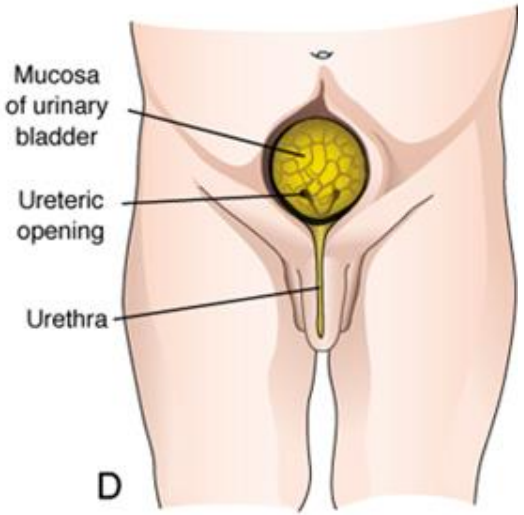
Penile Spadias

- Hypospadia
  - External urethral opening on the ventral surface
  - Underdeveloped penis
  - Curved ventrally
  - 3-5:1000 births
- Epispadia (fig. D)
  - Urethra opens along the dorsal surface of penis
  - 1:30,000
  - Often seen with exstrophy of bladder









Agenesis of External Genitalia

- Complete absence of external genitalia
- Very rare
- There is a urethral opening near anus



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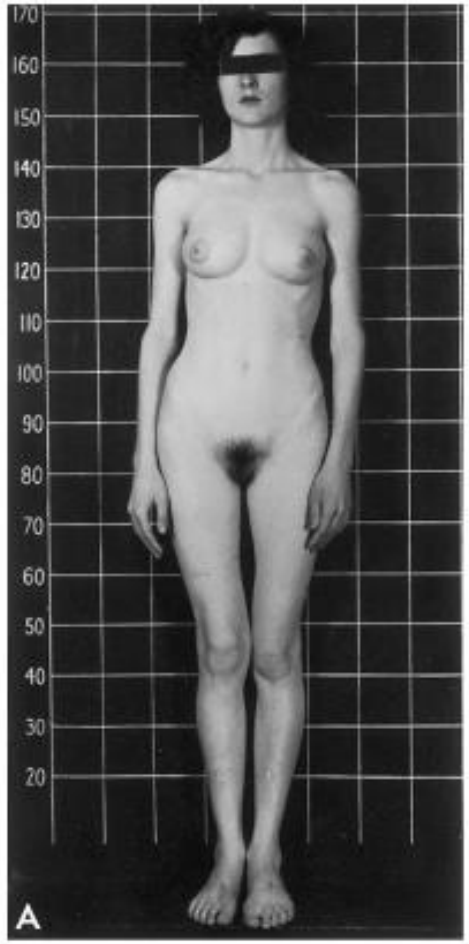
Differences in Sexual Development (DSD)

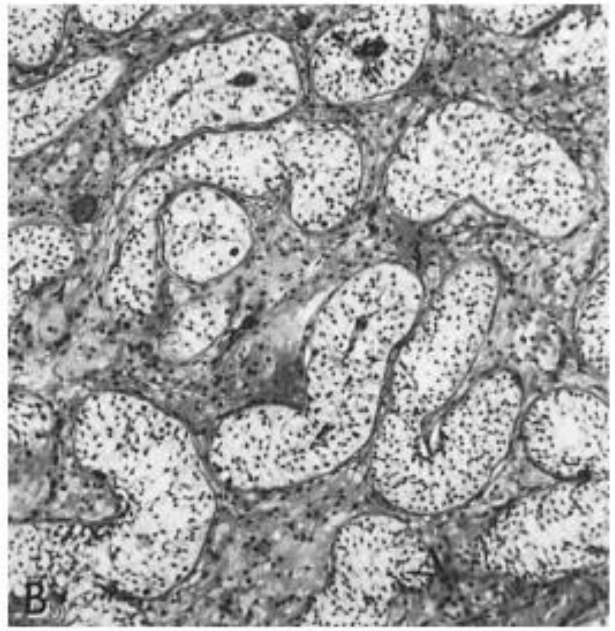
- Clitoral hypertrophy
  - XX genotype with an enlarged clitoris
- Micropenis
  - XY genotype with a penis size 2.5 deviations below the norm
- Intersexed individuals
  - Ovotestes - mixed gonadal tissues
  - 70% of affected are 46, XX with a uterus



Androgen Insensitivity Syndrome (AIS)

- Androgen insensitivity syndrome
  - Normal XY genotype
  - External genitalia often appears phenotypically XX
  - Undescended gonads
  - Often no fully formed uterus
  - Defect in the androgen receptor
- Can have partial or mild AIS
  - Mixed external genitalia
  - Partially descended testes





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Other Causes of DSD

- Congenital Adrenal Hyperplasia (CAH)
  - Most common form has deficiency in 21-hydroxylase
    - Decreased steroid hormone production
    - Increased ACTH
  - 46, XX with virilization (common cause of clitoral hypertrophy)
- 5-α-reductase deficiency
  - Affected 46, XY individuals have difficult to characterize genitalia

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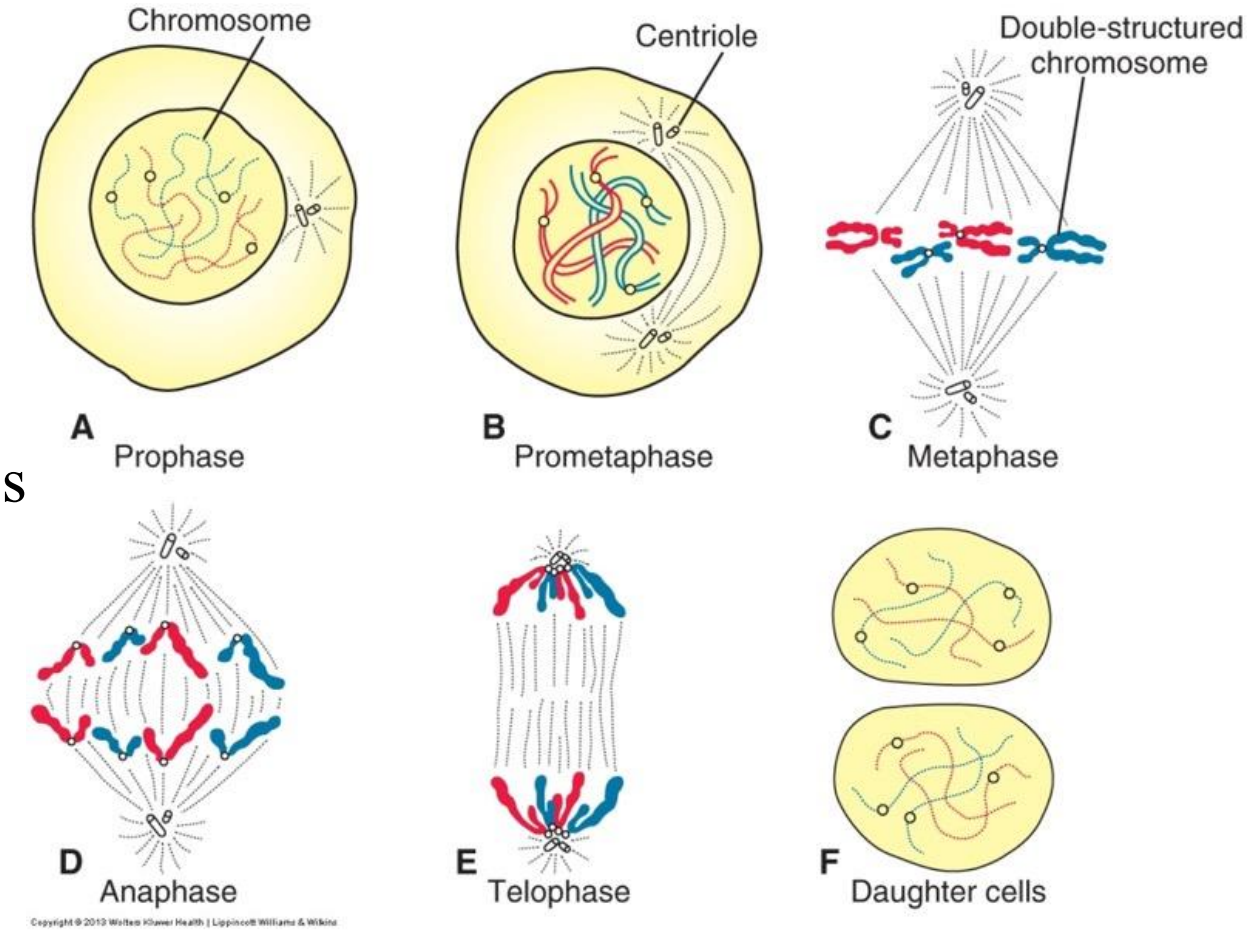
Development of Gametes

- Germs cells proliferate as they migrate and populated the gonadal cords
  - Mitosis maintains the diploid number of 46
- Gametogenesis requires meiosis
  - Reduction to 23 chromosomes

Mitosis

DNA is Duplicated

- Prophase – Nuclear membrane deconstructed and DNA condenses
- Metaphase – alignment of Chromosomes and Formation of spindle
- Anaphase- separation of chromosomes
- Telophase – re-establishment of Nucleus and division of cell



Meiosis

- Reduces the number of chromosomes
  - Two sequential divisions without an intervening round of DNA replication
- Ensures the production of a haploid phase of life
  - Gametes; Oocyte, Sperm
- Diploid state restored during fertilization

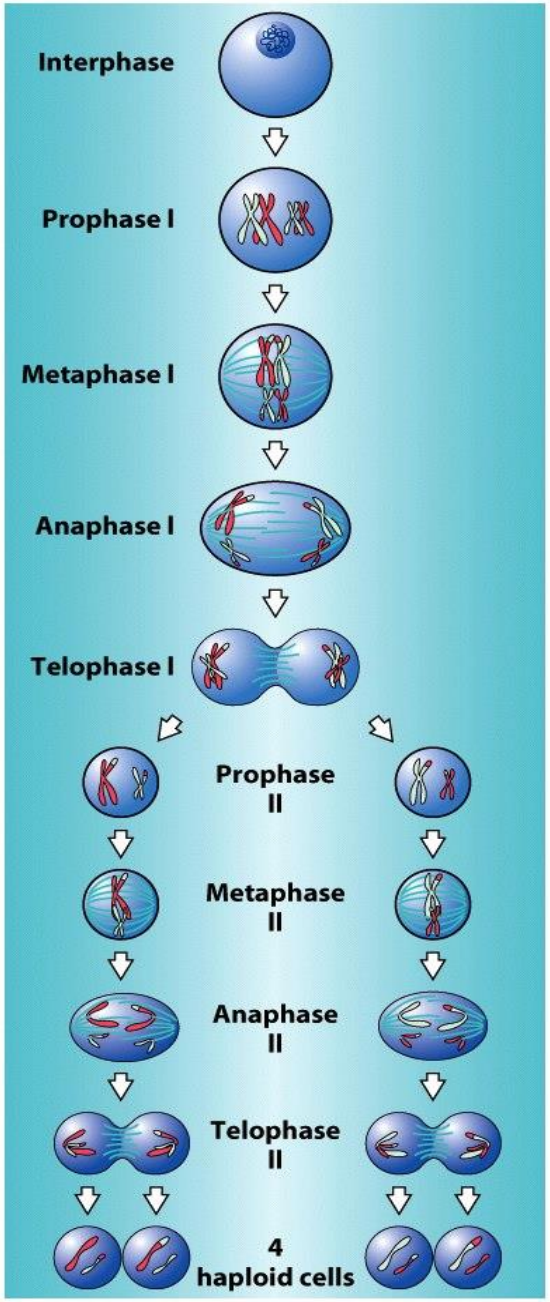


Figure 14-39 Cell and Molecular Biology, 5/e (© 2008 John Wiley & Sons)

Meiosis

- First division separates the homologous chromosome **pairs**
  - Prophase I – Telophase I
- Second division separates the sister chromatids
  - Prophase II – Telophase II
  - 4 haploid gametes

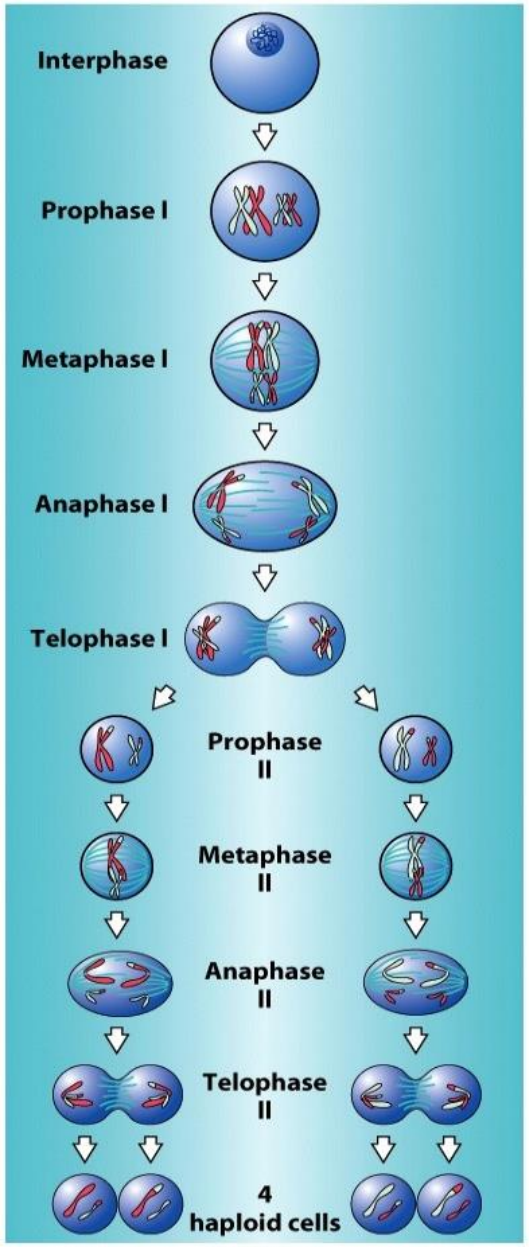
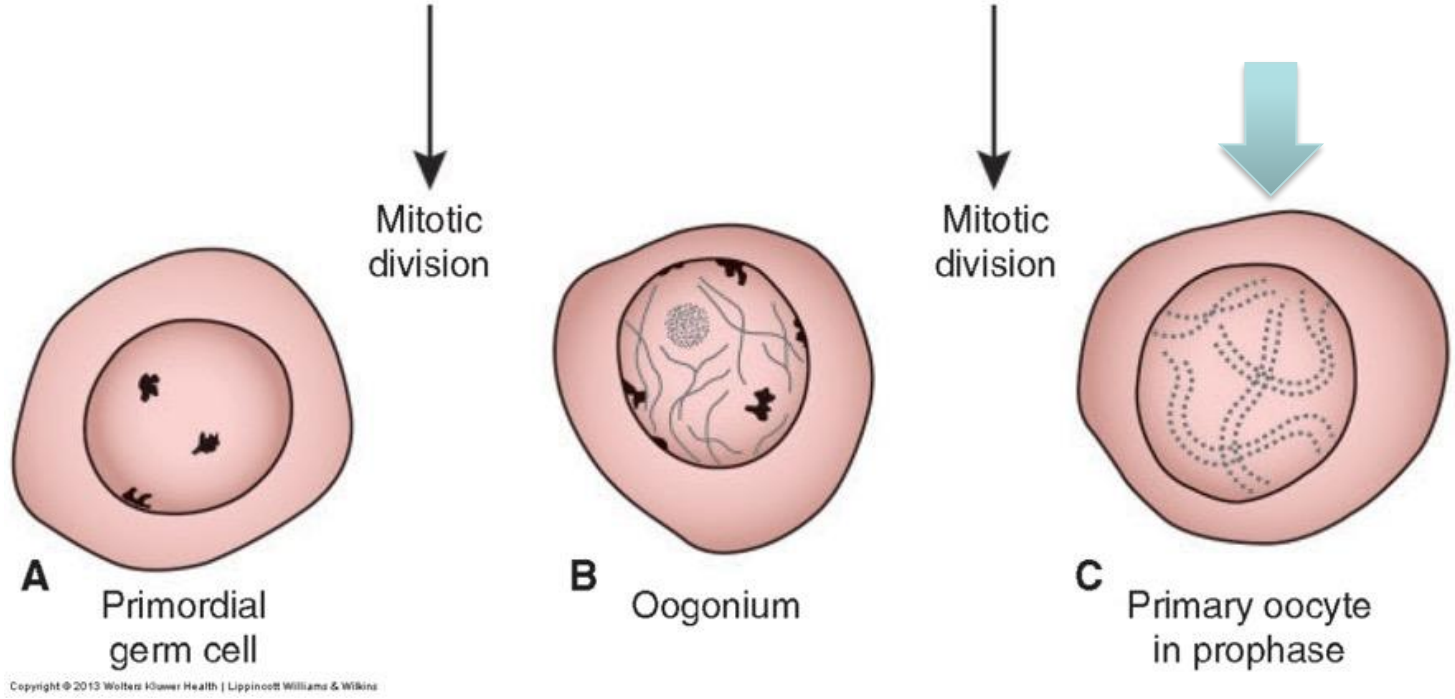


Figure 14-39 Cell and Molecular Biology, 5/e (©

Oogenesis: Primordial Germ Cells Enter Meiosis and Arrest as Primary Oocytes *Before* Birth

- Maturation of oocytes begins before birth - upon arrival of PGCs into the developing ovary.
- PGCs differentiate into oogonia and undergo many mitotic cell divisions.
- PGCs enter prophase of meiosis I and arrest as **primary oocytes organized as primordial follicles**.



Prophase I

- After DNA synthesis during Meiotic S-Phase the cells enter Prophase I
- Much longer than mitotic prophase
  - Many stages

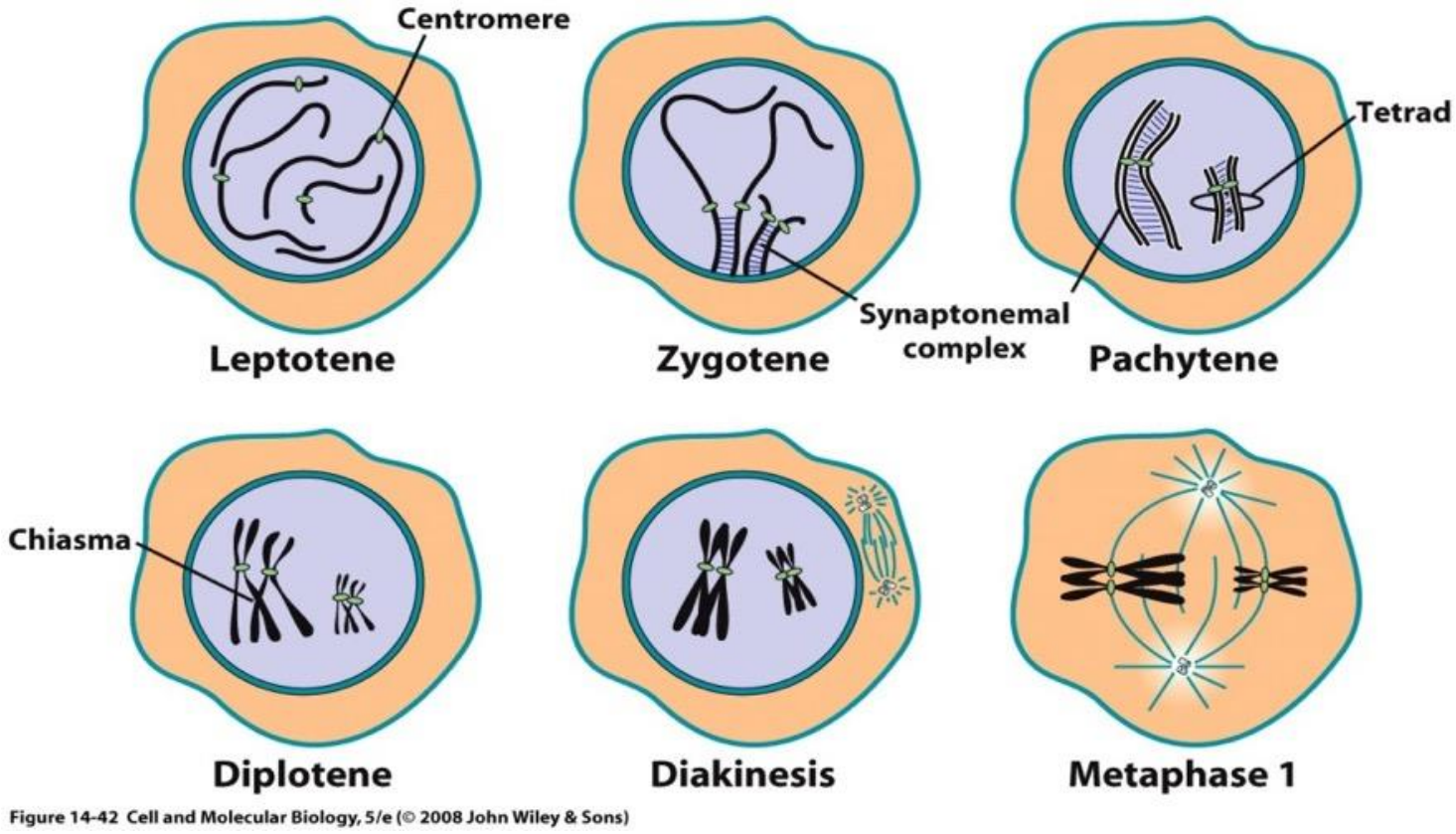


Figure 14-42 Cell and Molecular Biology, 5/e (© 2008 John Wiley & Sons)

Prophase I

- Leptotene – chromosomes visible
  - Sister chromatids not VISIBLY apparent but they are there
- Zygotene- homologous pairs distinctly aligned
  - Process of pairing called synapsis

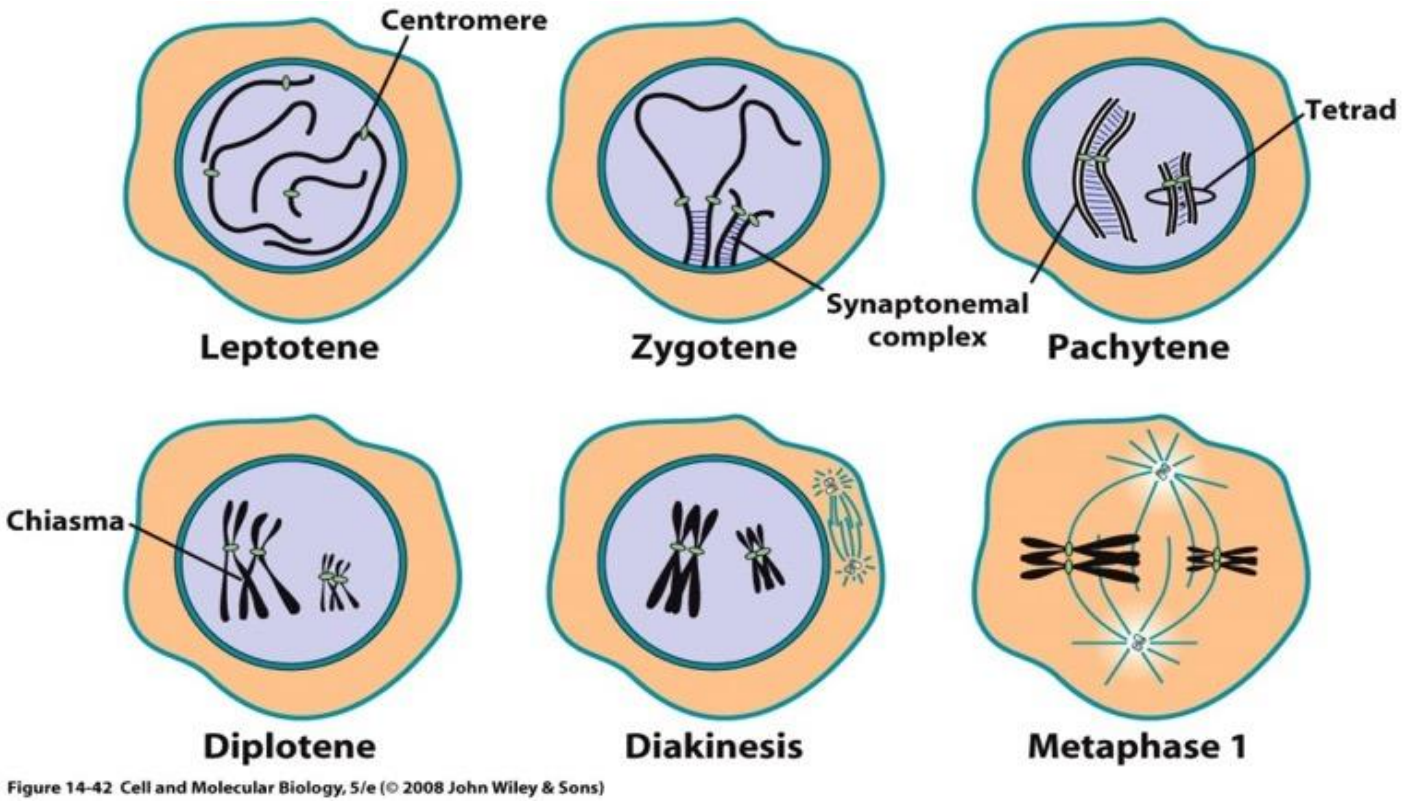


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Prophase I/Zygotene

- Synaptonemal complex formed during Zygotene
  - Ladder-like structure between the chromosome homologues
  - Homologues aligned along scaffold molecules
- Sister chromatids held together by cohesin
  - Aligned homologs called tetrads
- Complex can help genetic recombination

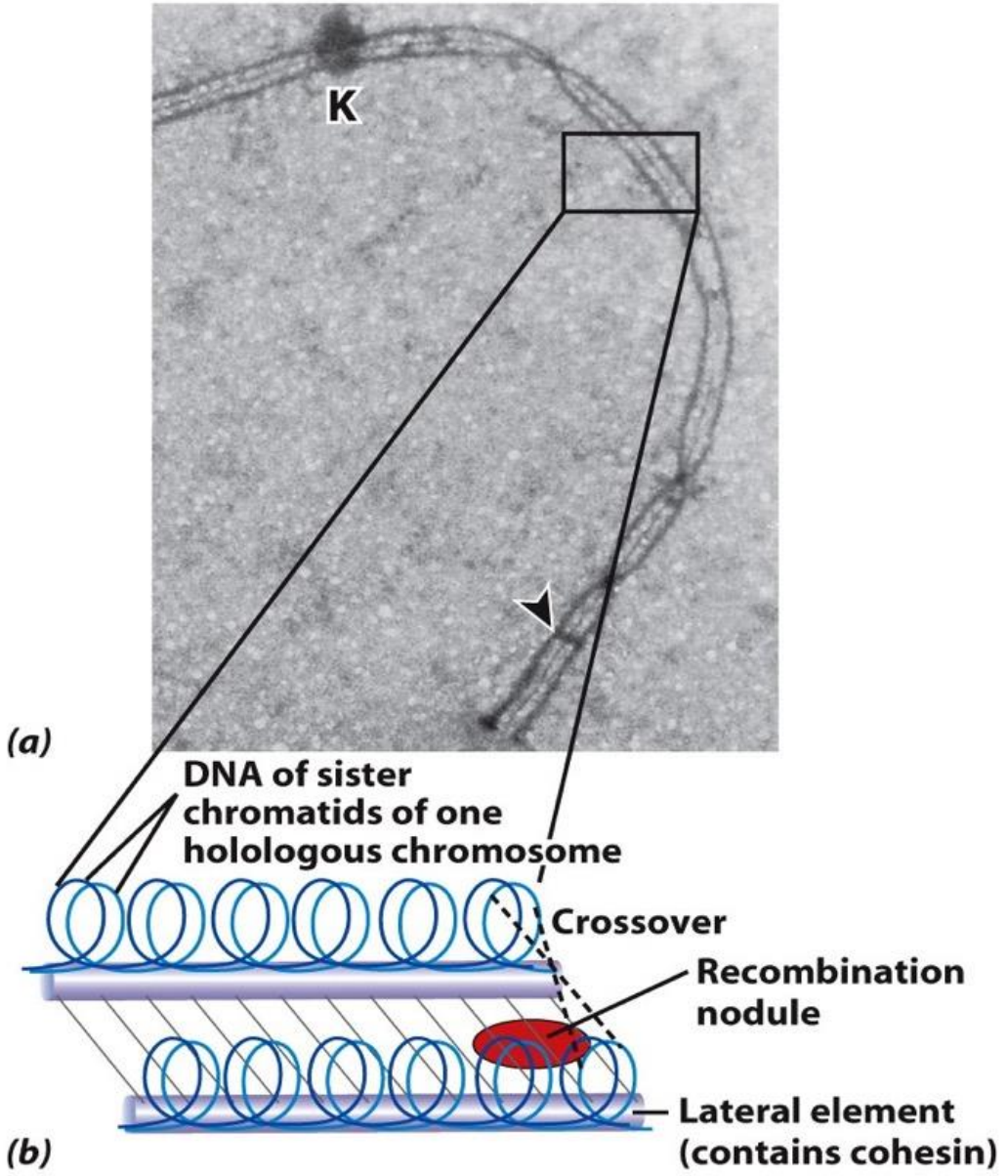


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Prophase I/ Pachytene

- Third stage of prophase I
- Electron dense regions called **recombination nodules** visible between the tetrads
  - Marks where crossing over will occur
  - Contain the enzymes needed to accomplish

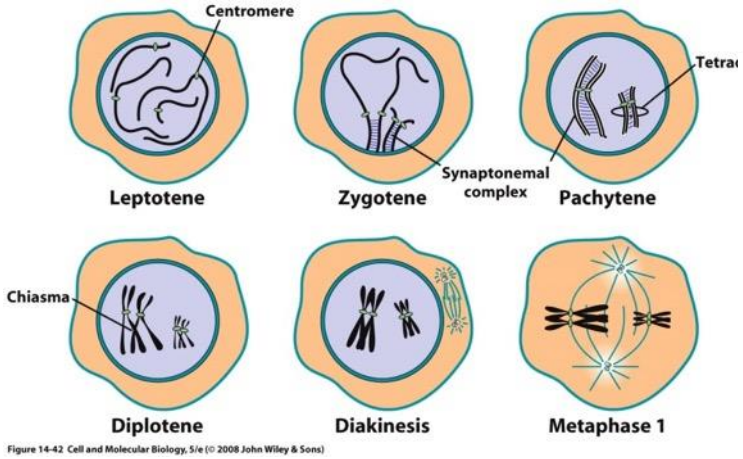


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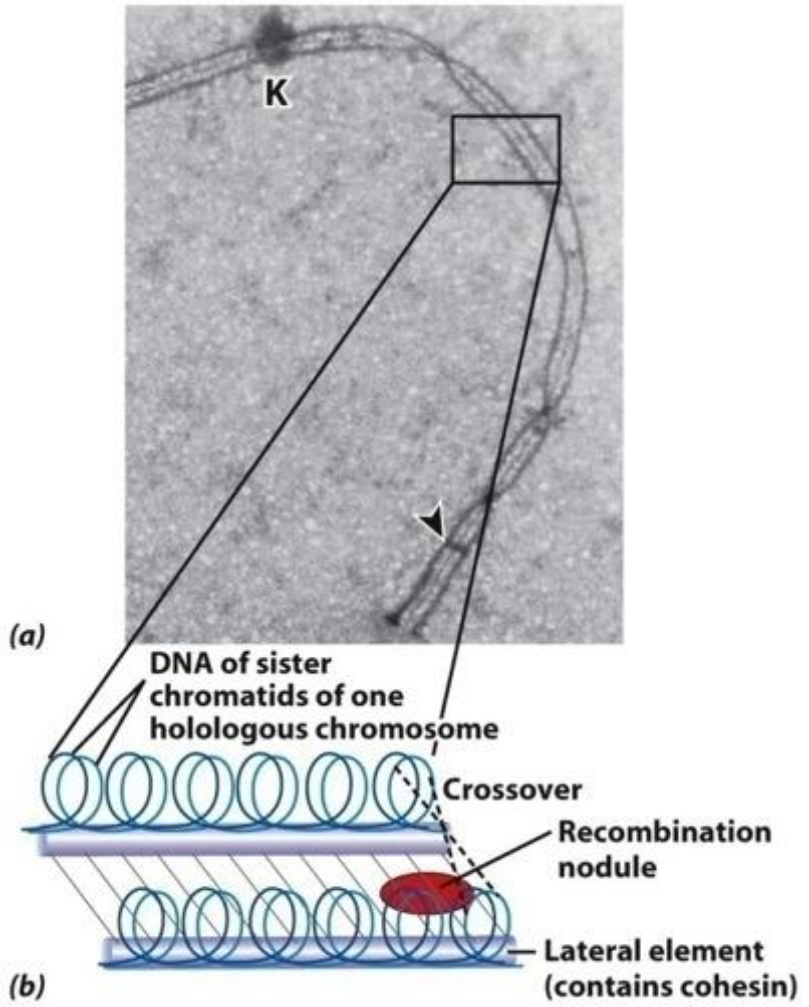


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Prophase I/ Diplotene

- Fourth stage of prophase I
  - Synaptonemal complex scaffold is breaking down??
- Visible evidence of cross over
  - Chiasmata
  - DNA strands move away from each other

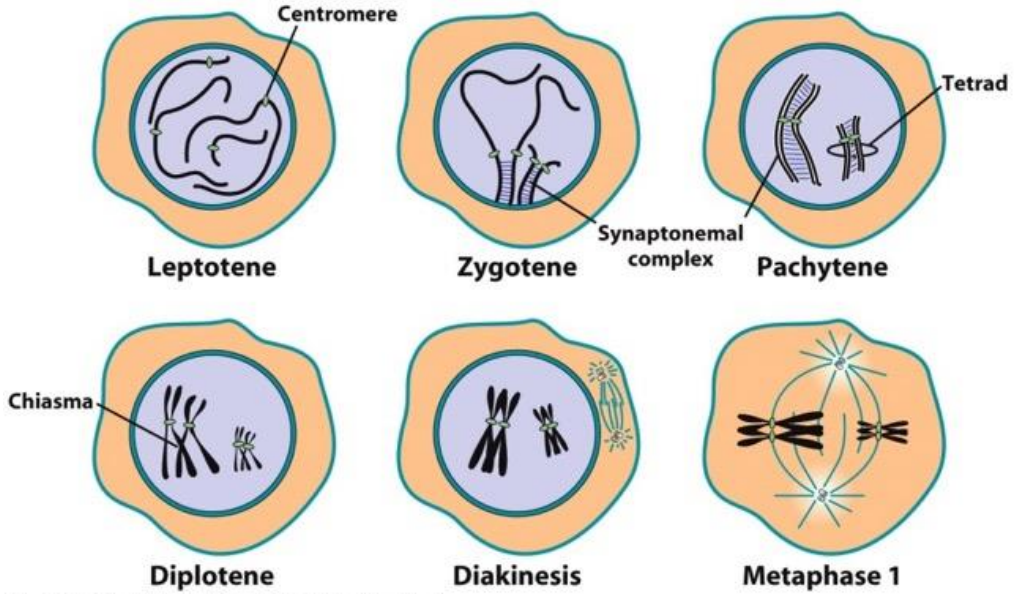
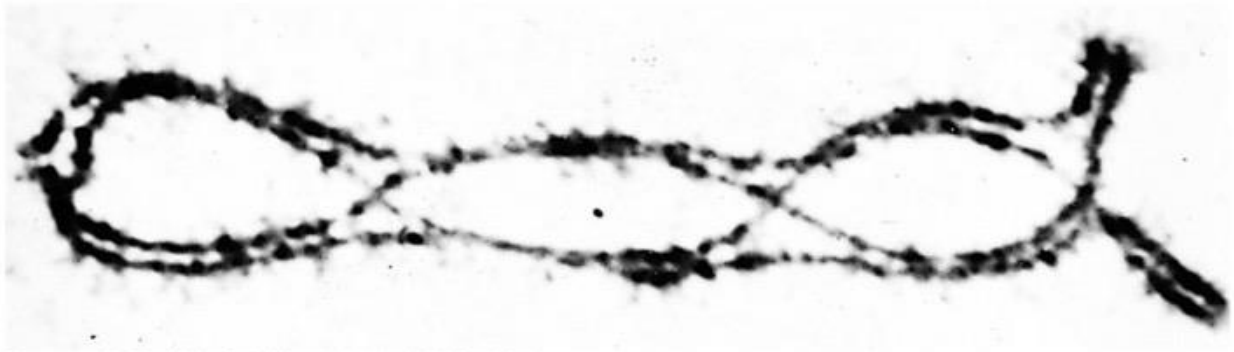
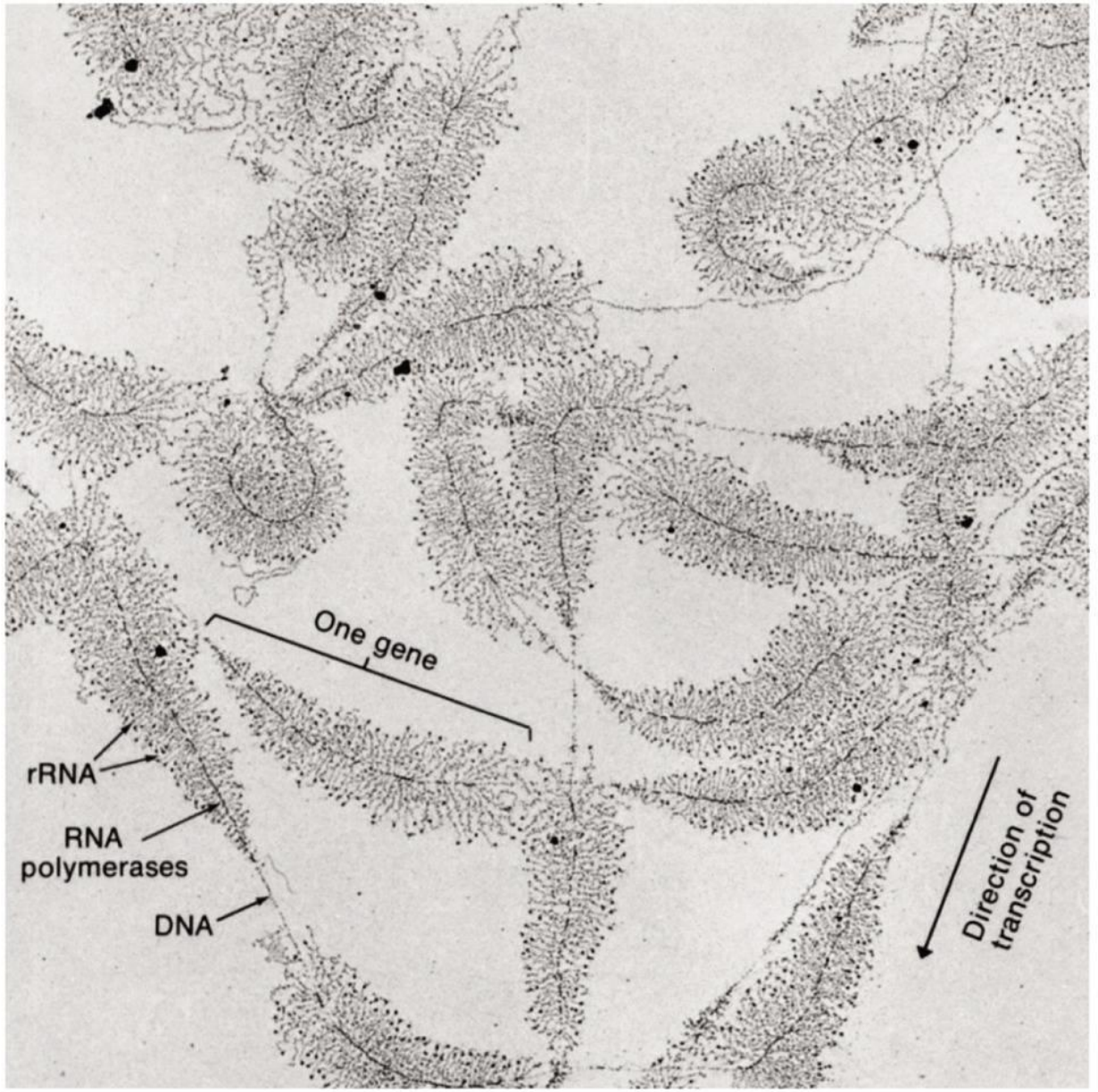


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Figure 14-45b Cell and Molecular Biology, 5/e (© 2008 John Wiley & Sons)

Diplotene in Oocyte

- Diplotene during oocyte formation is very long
- Also, intense metabolic activity
  - Oocyte prepping for fertilization in the future
- Areas of active transcription visible along the tetrad

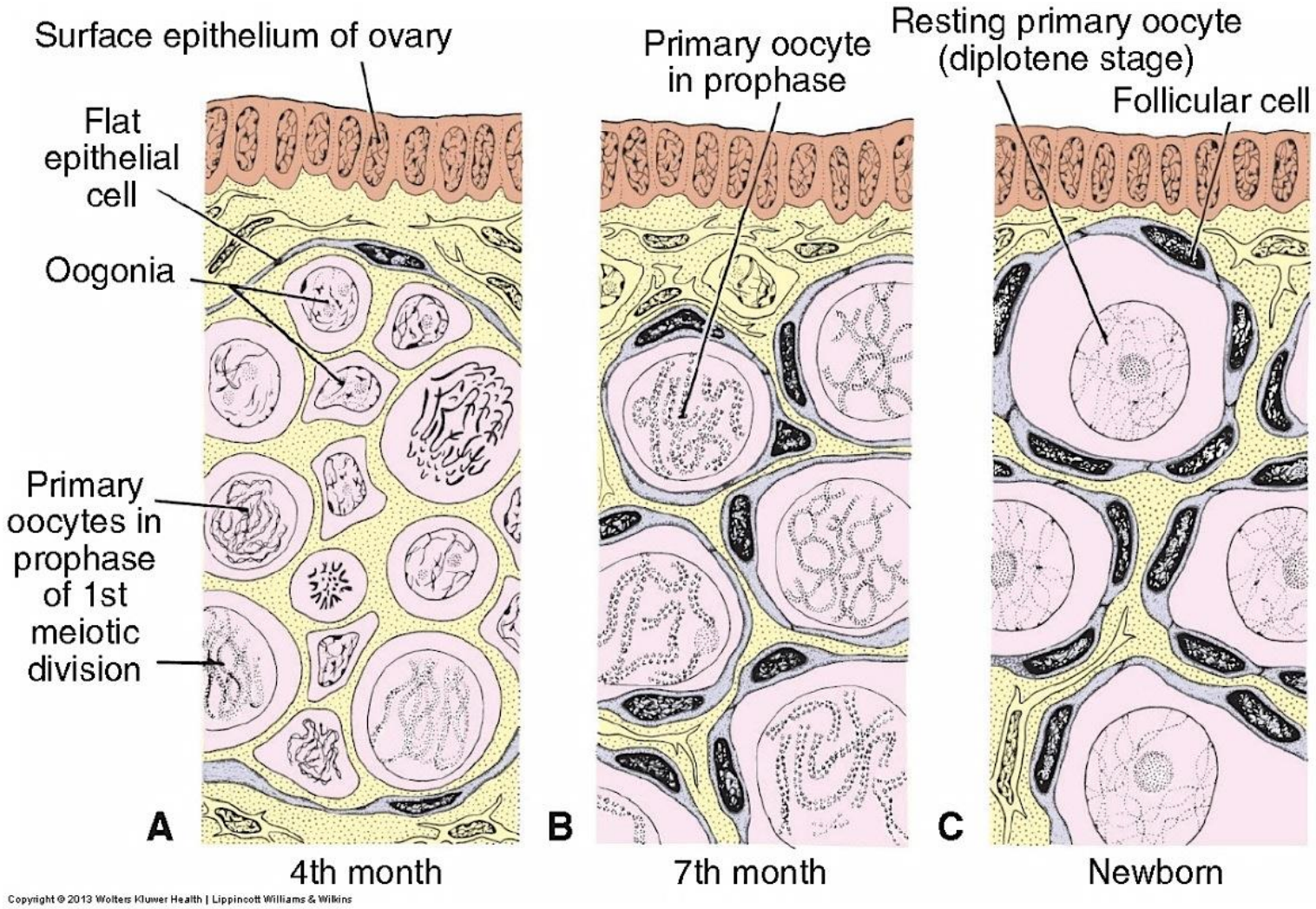


Oocyte Maturation & Ovulation

- 7 million germ cells by 5th month of fetal development
- Up to 2 million present at birth
- Less present at puberty
- Usually only one primary oocyte per ovarian cycle re-enters meiosis

  •400-500 mature to ovulation

  - Even less are produced if pregnancy occurs
  - Even less if birth control medication prevents ovulation



Diakinesis in Prophase I

- Diakinesis is the final stage of Prophase I
  - After sexual maturation and reactivation of development
- Spindle formation
- Homologous pairs are readied for separation

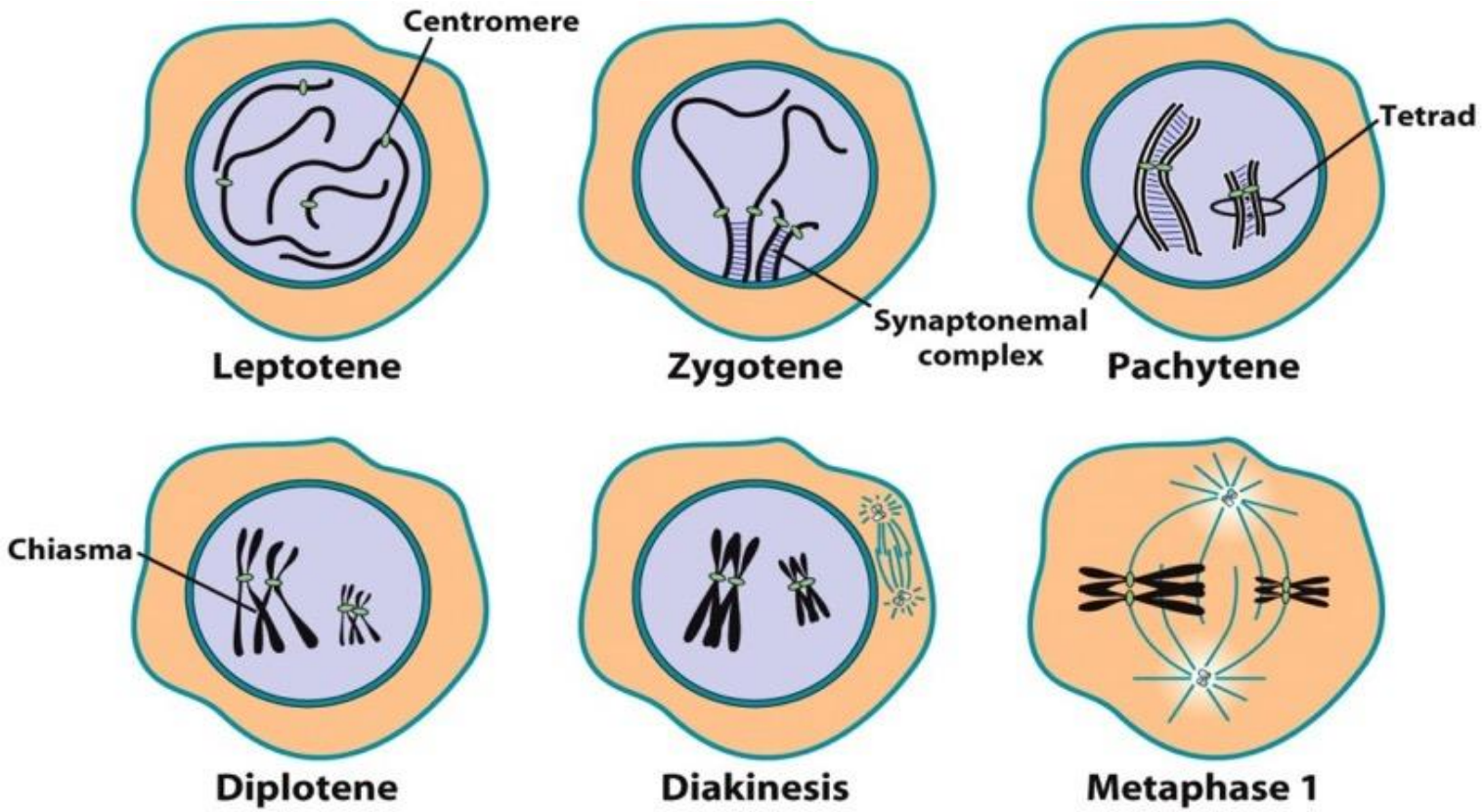
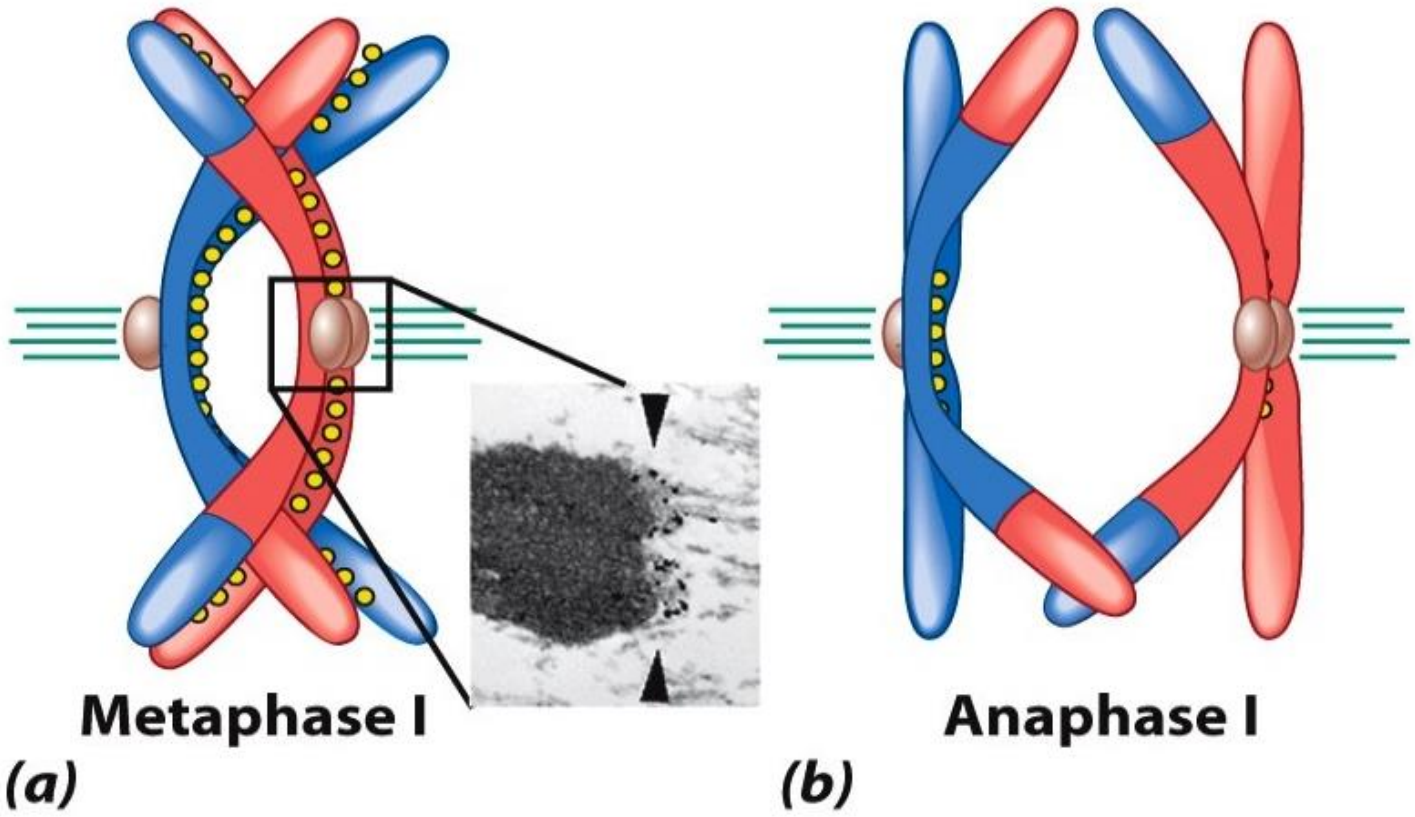


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Metaphase I

- During Metaphase I the homologous pairs are randomly aligned along midline
- Sister chromatids remain attached by cohesin at kinetochore
- Kinetochores of sisters aligned in same direction
  - This is so they can be captured by spindle from same side



Anaphase and Telophase I

- Anaphase I and Telophase I of Meiosis I not too dramatic
- Interkinesis phase separates Meiosis I from Meiosis II
  - Typically over a short time frame
- Prophase II much simpler than prophase I
  - More like a typical mitotic prophase

Metaphase II

- Kinetochores of sister chromatids aligned in opposite direction
- Human mature ***oocytes*** are arrested at Metaphase II when ovulation occurs
- Completion follow fertilization
  - Calcium influx at fertilization reactivates

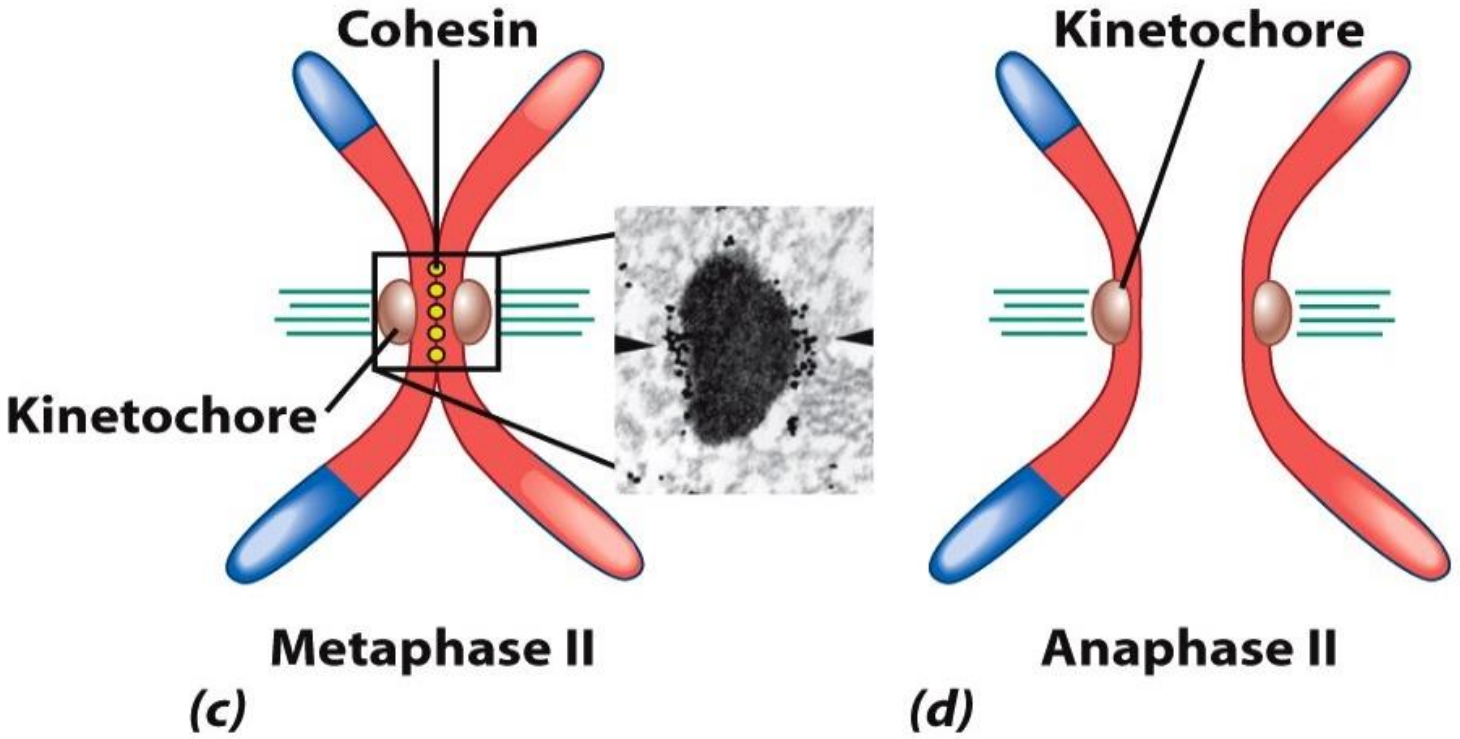


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Meiotic Errors and Chromosomal Abnormalities

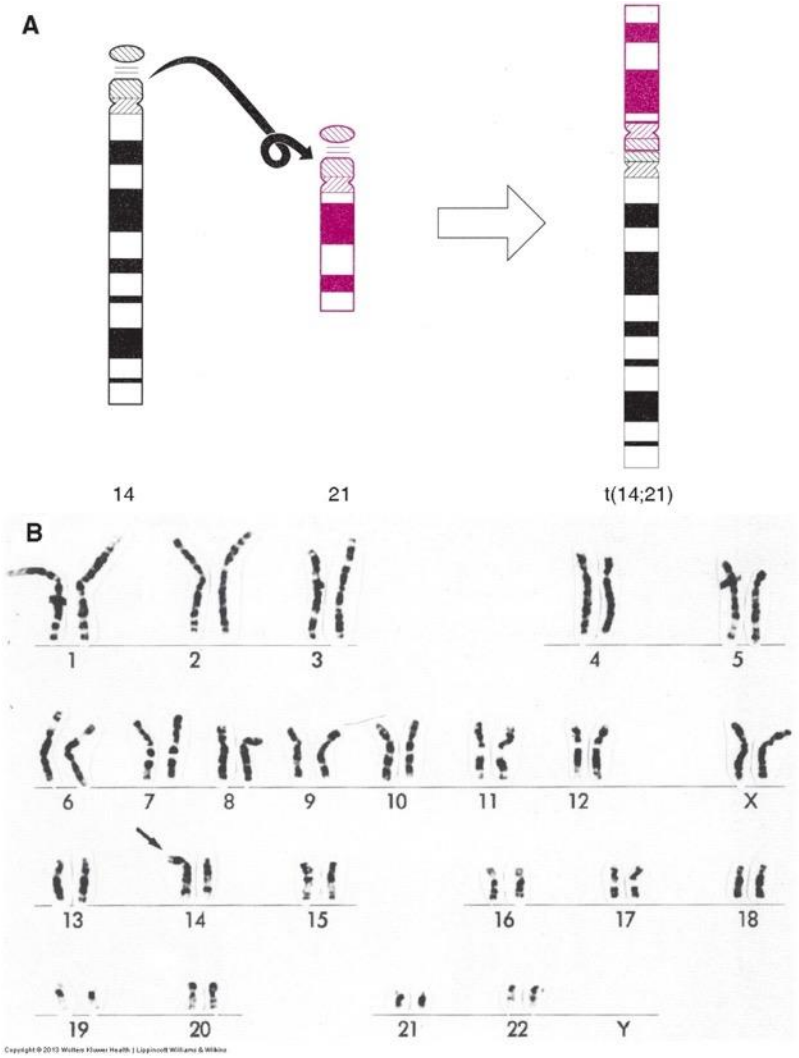
Chromosomal Translocation

- Exchange of genetic material due to crossing over between non-homologous chromosomes: variable clinical significance

Deletion

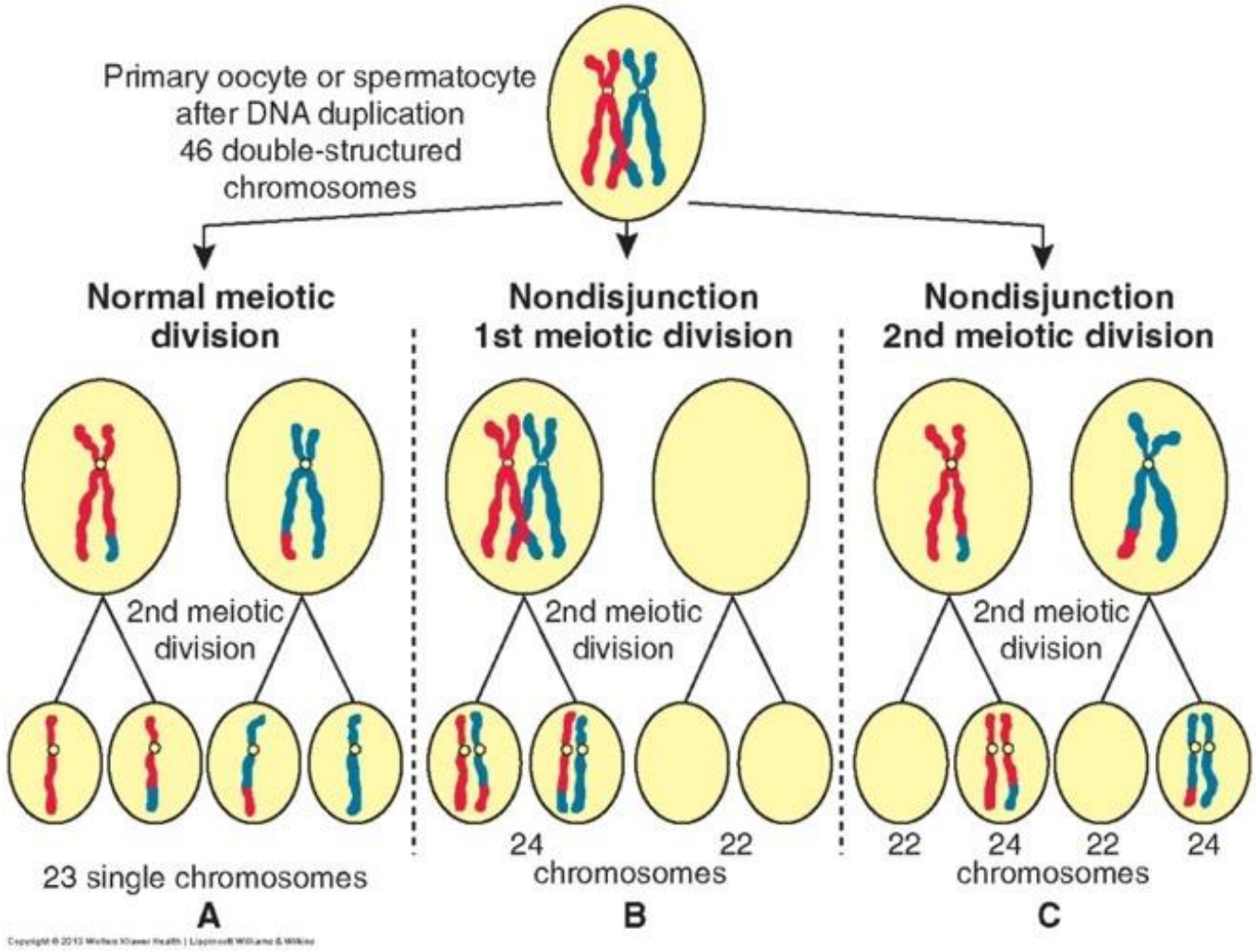
- Loss of a portion of a chromosome

Non-Disjunction



Non-Disjunction

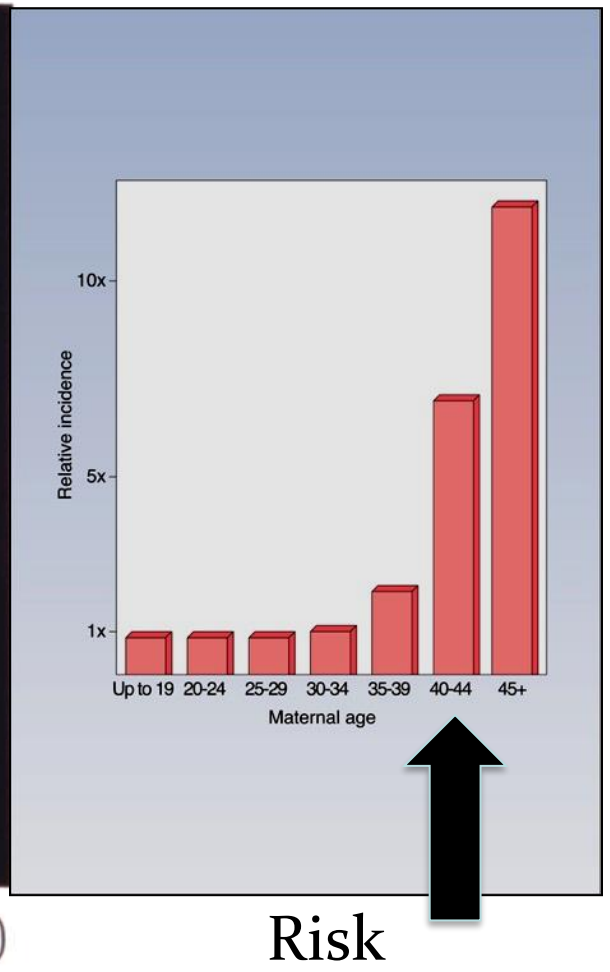
- Failure of separation
- Metaphase I affects the Homologous pairs splitting
- Metaphase II affects sister chromatids splitting



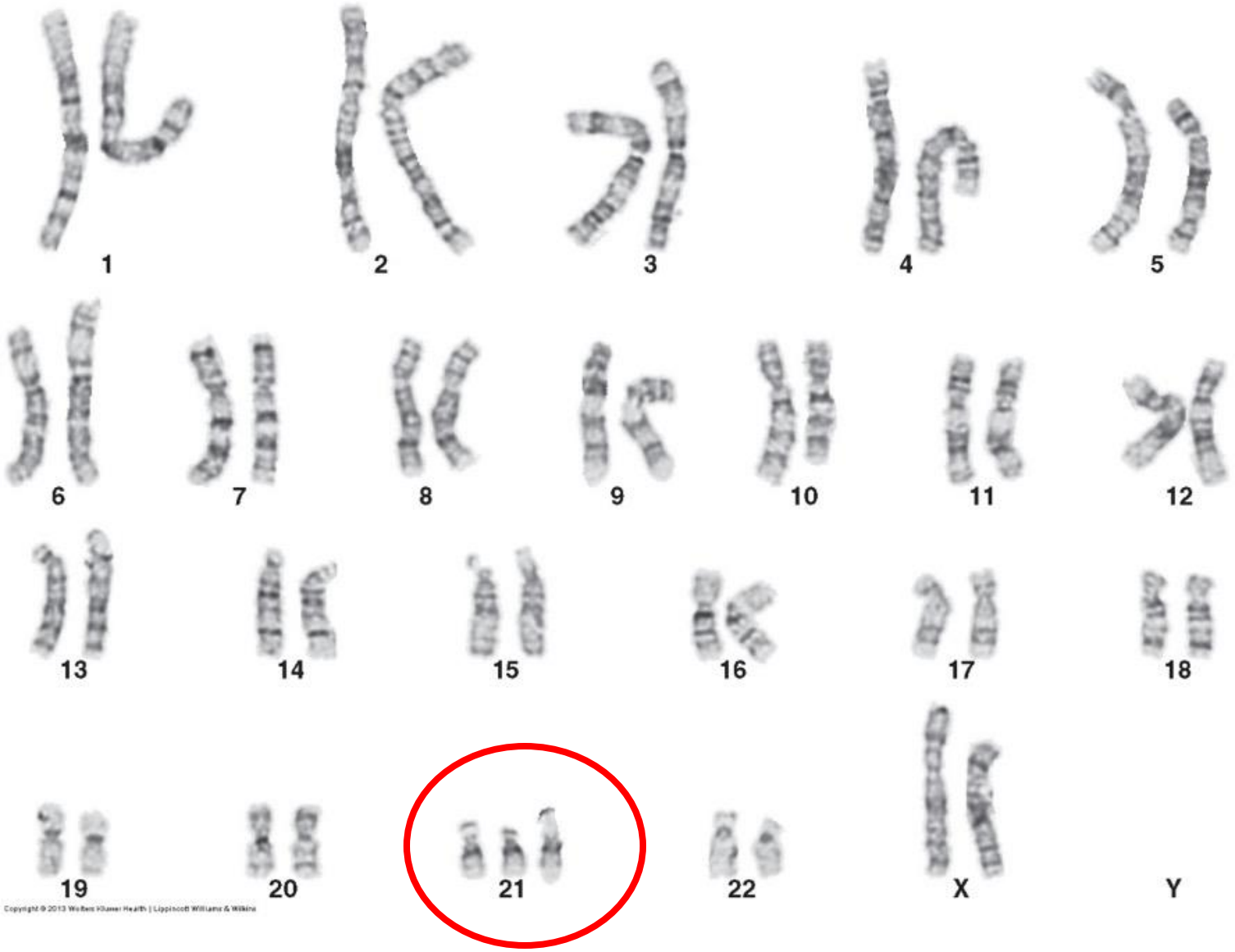
Chromosomal Nondisjunction



Misdirected. The older the oocyte, the more likely its chromosomes (red) will get stuck during meiosis, possibly due to faulty spindles (green).

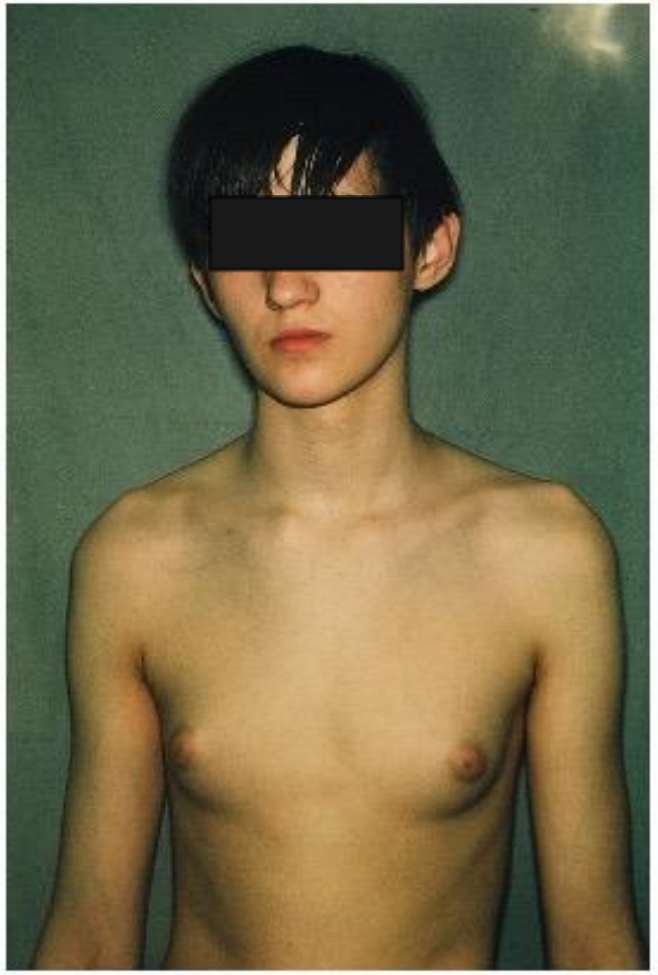


Karyotype of Trisomy 21: Down Syndrome



Hyperploidy of Sex Chromosomes

- XXX – individual
  - Normal ovaries, etc, fertile, mild intellectual disabilities in 25%
- XXY – individual
  - Hypogonadism (usually testis)
  - No sperm produced
  - Gynecomastia



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XXY individual; Klinefelter syndrome

Turner Syndrome

- 45, XO
  - One of most common chromosomal defects
  - 75% from paternal non-disjunction
- 1% survive
  - 1:8000 live births
- External genitalia includes clitoris, vagina and labia
- No 2nd sexual maturation



Summary of Cell Divisions and Cell Differentiation During Gametogenesis

Ploidy	Female germ cell line	Male germ cell line
Diploid	PGC	PGC
↓ Mitosis ↓		
Diploid	Oogonium	Spermatogonium
↓ Mitosis ↓		
Diploid	Primary oocyte (complete by birth; arrested in prophase of meiosis I until puberty, with one cell then proceeding to mature at each ovarian cycle)	Primary spermatocyte (spermatogenesis occurs continuously after puberty)
↓ Meiosis I ↓		
Haploid 2N	Seconday oocyte (released at ovulation; arrested in metaphase of meiosis II until fertilization)	Secondary spermatocyte
↓ Meiosis II ↓		
Haploid 1N	Mature oocyte or ovum (meiosis II completed upon fertilization)	Spermatid
↓		↓
First week of development Pronuclear fusion and zygote formation Cleavage and blastomere formation Compaction Morula Blastocyst Inner and outer cell mass (embryoblast and trophoblast) Uterine implantation		Morphologically mature spermatozoon (in seminiferous tubule) ↓ Biochemically mature spermatozoon (in epididymis) ↓ Fully functional spermatozoon (after capacitation in female reproductive tract)

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